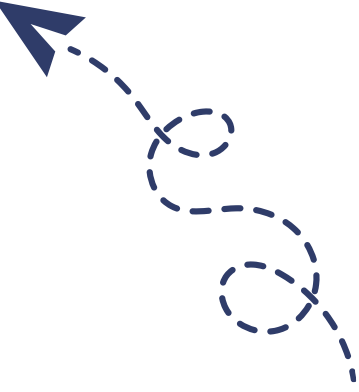




قناة تانية تمريضيانو على التليجرام





Medical-Surgical Nursing Practical Book

Nursing School – Second Year

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In order to support the Ministry of Health and Population in the delivery of education and training for nurses, the University of Sassari and the Italian Agency for Development Cooperation are proud to share the Nursing Curriculum Model for the first level of technical education.

The curriculum, which was developed in 2022, is a benchmark for all educators who are committed to providing students with skills that are up to international standards, in order to deliver high-quality care now and in the future.

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Medical Surgical Nursing Practical Part

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Capillary Blood Sample for Glucose Testing

Objectives:

Students will be able to:

- Define Capillary Blood Sample for Glucose Testing
 - Identify the goal of Capillary Blood Sample for Glucose Testing
 - Determine equipment's needed during the procedure
 - Demonstrate Capillary Blood Sample for Glucose Testing procedure
- Blood glucose monitoring allows patients with diabetes mellitus to self-manage their disease. Obtaining capillary blood by skin puncture is less painful than venipuncture, and the ease of the skin puncture method makes it possible for patients to perform this procedure.
- **The goal** of this test is the Patients with diabetes use this type of testing as an important part of disease management to monitor blood glucose and adjust lifestyle interventions and treatment

➤ Equipment

- Blood glucose meter
- Sterile lancet
- Cotton balls or gauze squares
- Nonsterile gloves
- Additional PPE, as indicated
- Skin cleanser and water or alcohol swab
- Testing strips for meter

➤ Procedure

Steps	Rational
1. Gather equipment. Check expiration date on blood test strips.	This provides an organized approach to the task. Blood test strips that are past expiration date could cause inaccurate results and should not be used.
2. Perform hand hygiene and put on PPE, if indicated.	Hand hygiene and PPE prevent the transmission of microorganisms. PPE is required based on transmission precautions.
3. Identify the patient. Explain the procedure to the patient and instruct the patient about the need for monitoring blood glucose	Identifying the patient ensures the right patient receives the intervention and helps prevent errors. Explanation helps to alleviate

	anxiety and facilitate cooperation.
4. Close curtains around the bed and close the door to the room, if possible.	Closing the curtain or door provides for patient privacy
5. Prepare lancet using aseptic technique.	Aseptic technique maintains sterility
6. Remove test strip from the vial. Recap container immediately. Test strips also come individually wrapped. Check that the code number for the strip matches the code number on the monitor screen.	Immediate recapping protects strips from exposure to humidity, light, and discoloration. Matching code numbers on the strip and glucose monitor ensures that the machine is calibrated correctly.
7. Insert the strip into the meter according to directions for that specific device. Alternately, strip may be placed in meter after collection of samples on test strip, depending on meter in use.	Correctly inserted strip allows meter to read blood glucose level accurately.
	
8. Cleanse the skin with an alcohol swab. Allow skin to dry completely.	Washing with skin cleanser and water or alcohol cleanses the puncture site. Warm water also helps to cause vasodilation. Alcohol can interfere with accuracy of results if not completely dried
9. Hold lancet perpendicular to skin and pierce site with lancet	Holding lancet in proper position facilitates proper skin penetration.
	
10. Wipe away first drop of blood with gauze square or cotton ball if recommended by manufacturer of monitor	Manufacturers recommend discarding the first drop of blood, which may be contaminated by

	serum or cleansing product, producing an inaccurate reading.
11. Encourage bleeding by lowering the hand, making use of gravity. Lightly stroke the finger, if necessary, until a sufficient amount of blood has formed to cover the sample area on the strip	An appropriate-sized droplet facilitates accurate test results.
12. Gently touch a drop of blood to the test strip without smearing it	Smearing blood on the strip may result in inaccurate test results
	
13. Apply pressure to puncture site with a cotton ball or dry gauze. Do not use alcohol wipe.	Pressure causes hemostasis. Alcohol stings and may prolong bleeding.
14. Read blood glucose results and document appropriately at bedside. Inform patient of test result.	Timing depends on type of meter.
15. Turn off meter, remove test strip, and dispose of supplies appropriately. Place lancet in sharps container.	Proper disposal prevents exposure to blood and accidental needlestick.
16. Remove gloves and any other PPE, if used. Perform hand hygiene.	Removing PPE properly reduces the risk for infection transmission and contamination of other items. Hand hygiene reduces the transmission of microorganisms.

➤ Procedure Check List

PROCEDURE STEPS	Mark	Trial 1	Trial 2	Trial 3	comment
1. Gather equipment.					
2. Perform hand hygiene and put on PPE, if indicated					
3. Identify the patient. Explain the procedure to the patient and instruct the patient about the need for monitoring blood glucose					
4. Close curtains around bed and close the door to the room, if possible					
5. Prepare lancet using aseptic technique					
6. Remove test strip from the vial. Recap container immediately. Check that the code number for the strip matches code number on the monitor screen.					
7. Insert the strip into the meter according to directions for that specific device.					
8. cleanse the skin with an alcohol swab. Allow skin to dry completely.					
9. Hold lancet perpendicular to skin and pierce site with lancet					
10. Wipe away first drop of blood with gauze square or cotton ball if recommended by manufacturer of monitor					
11. Encourage bleeding by lowering the hand, making use of gravity. Lightly stroke the finger, if necessary, until a sufficient amount of blood has formed to cover the sample area on the strip					
12. Gently touch a drop of blood to the test strip without smearing it					
13. Apply pressure to puncture site with a cotton ball or dry gauze. Do not use alcohol wipe.					
14. Read blood glucose results and document appropriately at bedside. Inform patient of test result.					
15. Turn off meter, remove test strip, and dispose of supplies appropriately. Place lancet in sharps container.					
16. Remove gloves and any other PPE, if used. Perform hand hygiene.					

Peripheral Intravenous Care

Objectives:

Students will be able to:

- Identify the goal of Peripheral Intravenous Care
 - Determine equipment's needed during the procedure
 - Demonstrate Peripheral Intravenous Care procedure
-
- The IV site is a potential entry point for microorganisms into the bloodstream. To prevent this, sealed IV dressings are used to occlude the site and prevent complications.
 - Whenever these dressings need to be changed, it is important to observe meticulous aseptic technique to minimize the possibility of contamination.

➤ Equipment

- Transparent occlusive dressing
- 2% chlorhexidine, povidone-iodine, 70% alcohol
- Adhesive remover (optional)
- Alcohol swabs
- Tape
- Clean gloves
- Towel or disposable pad
- Additional PPE, as indicated

➤ Procedure

Steps	Rational
1. Determine the need for a dressing change. Check facility policy. Gather all equipment and bring to bedside.	Immediately change any access site dressing that is damp, loosened, or soiled. Having equipment available saves time and facilitates accomplishment of procedure.
2. Perform hand hygiene and put on PPE, if indicated.	Hand hygiene and PPE prevent the spread of microorganisms. PPE is required based on transmission precautions
3. Identify the patient	Identifying the patient ensures the right patient receives the intervention and helps prevent errors

4. Close curtains around bed and close the door to the room, if possible. Explain what you are going to do and why you are going to do it to the patient. Ask the patient about allergies to tape and skin antiseptics.	This ensures the patient's privacy. Explanation relieves anxiety and facilitates cooperation. Possible allergies may exist related to tape or antiseptics
5. Put on mask and place a mask on patient, if indicated. Put on gloves. Place towel or disposable pad under the arm with the venous access. If solution is currently infusing, temporarily stop the infusion. Hold the catheter in place with your nondominant hand and carefully remove old dressing and/or stabilization/securing device. Use adhesive remover as necessary. Discard dressing 	Masks should be used for catheter with extended dwell time or if patient is immunocompromised. Gloves prevent contact with blood and body fluids. Pad protects underlying surface. Proper disposal of dressing prevents transmission of microorganisms.
6. Inspect IV site for presence of phlebitis (inflammation), infection, or infiltration. Discontinue and relocate IV, if noted.	Inflammation (phlebitis), infection, or infiltration causes trauma to tissues and necessitates removal of the venous access device.
7. Cleanse site with an antiseptic solution such as chlorhexidine or according to facility policy. apply chlorhexidine using a back-and-forth friction scrub for at least 30 seconds. Do not wipe or blot. Allow to dry completely.	Scrubbing motion and length of time (minimum 30 seconds) is necessary for chlorhexidine to be effective
8. Place sterile transparent dressing over venipuncture site 	Transparent dressing allows easy visualization and protects the site.

9. Label dressing with date, time of change, and initials. Loop the tubing near the entry site, and anchor with tape (nonallergenic) close to site 	Labeling helps ensure communication about venous access site dressing change. The weight of the tubing is sufficient to pull it out of the vein if it is not well anchored.
10. Remove equipment. Ensure patient's comfort. Remove gloves. Lower bed, if not in lowest position.	Promotes patient comfort and safety. Removing gloves properly reduces the risk for infection transmission and contamination of other items.
11. Remove additional PPE, if used. Perform hand hygiene.	Removing PPE properly reduces the risk for infection transmission and contamination of other items. Hand hygiene prevents transmission of microorganisms.

➤ Procedure Check List

PROCEDURE STEPS	Mark	Trial 1	Trial 2	Trial 3	Comment
1. Determine the need for a dressing change. Check facility policy. Gather all equipment and bring to bedside.					
2. Perform hand hygiene and put on PPE, if indicated.					
3. Identify the patient					
4. Close curtains around bed and close the door to the room, if possible. Explain what you are going to do to the patient. Ask the patient about allergies to tape and skin antiseptics.					
5. Put on mask and place a mask on patient, if indicated. Put on gloves. Place towel or disposable pad under the arm with the venous access. If solution is currently infusing, temporarily stop the infusion. Hold the catheter in place with your nondominant hand and carefully remove old dressing and/or stabilization/securing device. Use adhesive remover as necessary. Discard dressing					
6. Inspect IV site for presence of phlebitis (inflammation), infection, or infiltration. Discontinue and relocate IV, if noted.					
7. Cleanse site with an antiseptic solution such as chlorhexidine or according to facility policy. apply chlorhexidine using a back-and-forth friction scrub for at least 30 seconds. Do not wipe or blot. Allow to dry completely.					
8. Place sterile transparent dressing over venipuncture site					
9. Label dressing with date, time of change, and initials. Loop the tubing near the entry site, and anchor with tape (nonallergenic) close to site					
10. Remove equipment. Ensure patient's comfort. Remove gloves. Lower bed, if not in lowest position.					
11. Remove additional PPE, if used. Perform hand hygiene.					

Cleaning a Wound and Applying a Dry, Sterile Dressing

Objectives:

Students will be able to:

- Define Cleaning a Wound
 - Identify the goal of Cleaning a Wound and Applying a Dry, Sterile Dressing
 - Determine equipment's needed during the procedure
 - Demonstrate Cleaning a Wound and Applying a Dry, Sterile Dressing procedure
- **Wound cleaning** is performed to remove debris, contaminants, and excess exudate. Sterile normal saline or a commercially prepared cleanser is the preferred cleansing solution.
- **The goal** of wound care is to promote tissue repair and regeneration to restore skin integrity. Often, wound care includes cleaning of the wound and the use of a dressing as a protective covering over the wound.
- There is no standard frequency for how often dressings should be changed. It depends on the amount of drainage, the primary practitioner's preference, the nature of the wound, and the particular wound care product being used.

➤ Equipment

- Sterile gloves
- Clean, disposable gloves
- Additional PPE, as indicated
- Gauze dressings
- Surgical or abdominal pads
- Sterile dressing set or suture set (for the sterile scissors and forceps)
- Sterile cleaning solution as order (commonly 0.9% normal saline solution, or a commercially prepared wound cleanser)
- Sterile basin (may be optional)
- Sterile drape (may be optional)
- Plastic bag or other appropriate waste container for soiled dressings
- Waterproof pad and bath blanket
- Tape or ties
- Bath blanket or other linens for draping patient
- Additional dressings and supplies needed or required by the primary care provider's order

➤ Procedure

Steps	Rational
1. Review the medical order for wound care or the nursing plan of care related to wound care. Gather necessary supplies	Reviewing the order and plan of care validates the correct patient and correct procedure. Preparation promotes efficient time management and an organized approach to the task.
2. Perform hand hygiene and put on PPE, if indicated	Hand hygiene and PPE prevent the spread of microorganisms. PPE is required based on transmission precautions.
3. Identify the patient.	Identifying the patient ensures the right patient receives the intervention and helps prevent errors.
4. Assemble equipment on overbed table within reach.	Organization facilitates performance of the task.
5. Close the curtains around the bed and close the door to the room, if possible. Explain to the patient what you are going to do	This ensures the patient's privacy. Explanation relieves anxiety and facilitates cooperation.
6. Assess the patient for the possible need for nonpharmacologic pain-reducing interventions or analgesic medication before wound care dressing change. Administer appropriate prescribed analgesic. Allow enough time for the analgesic to achieve its effectiveness.	Pain is a subjective experience influenced by past experience. Wound care and dressing changes may cause pain for some patients.
7. Assist the patient to a comfortable position that provides easy access to the wound area.	Patient positioning and use of a bath blanket provide for comfort and warmth.
8. Check the position of drains, tubes, or other adjuncts before removing the dressing. Put on clean, disposable gloves and loosen tape on the old dressings	Checking ensures that a drain is not removed accidentally if one is present. Gloves protect the nurse from contaminated dressings and prevent the spread of microorganisms.



9. Carefully remove the soiled dressings



If any part of the dressing sticks to the underlying skin, use small amounts of sterile saline to help loosen and remove it



Cautious removal of the dressing is more comfortable for the patient and ensures that any drain present is not removed.

Sterile saline moistens the dressing for easier removal and minimizes damage and pain.

After removing the dressing, note the presence, amount, type, color, and odor of any drainage on the dressings



Place soiled dressings in the appropriate waste receptacle. Remove your gloves and dispose of them in an appropriate waste receptacle



The presence of drainage should be documented.

Proper disposal of soiled dressings and used gloves prevents the spread of microorganisms.

10. Inspect the wound site for size, appearance, and drainage. Assess if any pain is present. Check the status of sutures, adhesive closure strips, staples, and drains or tubes, if present. Note any problems to include in your documentation.

Wound healing or the presence of irritation or infection should be documented.

11. Using sterile technique, prepare a sterile work area and open the needed supplies

Supplies are within easy reach and sterility is maintained.

12. Put on sterile gloves



Use of sterile gloves maintains surgical asepsis and sterile technique and reduces the risk for spreading microorganisms.

13. Clean the wound. Clean the wound from top to bottom and from the center to the outside. Following this pattern, use new gauze for each wipe, placing the used gauze in the waste receptacle



Cleaning from top to bottom and center to outside ensures that cleaning occurs from the least to most contaminated area and a previously cleaned area is not contaminated again.

14. Once the wound is cleaned, dry the area using a gauze sponge in the same manner.



Moisture provides a medium for growth of microorganisms.

15. Apply a layer of dry, sterile dressing over the wound. Forceps may be used to apply the dressing.



Primary dressing serves as a wick for drainage. Use of forceps helps ensure that sterile technique is maintained.

16. Apply a surgical or abdominal pad (ABD) over the gauze at the site of the outermost layer of the dressing, as necessary



The dressing acts as additional protection for the wound against microorganisms in the environment and increased absorption of drainage.

17. Remove and discard gloves. Apply tape, Montgomery straps, or roller gauze to secure the dressings.

Proper disposal of gloves prevents the spread of microorganisms. Tape or other securing products are easier to apply after gloves have been removed.

18. After securing the dressing, label it with date and time. Remove all remaining equipment; place the patient in a comfortable position, with side rails up and bed in the lowest position.

Recording date and time provides communication and demonstrates adherence to plan of care. Proper patient and bed positioning promotes safety and comfort.

19. Remove PPE, if used. Perform hand hygiene.

Removing PPE properly reduces the risk for infection transmission and contamination of other items. Hand hygiene prevents the spread of microorganisms

➤ Procedure Check List

PROCEDURE STEPS	Mark	Trial 1	Trial 2	Trial 3	Comment
1. Review the medical order for wound care or the nursing plan of care related to wound care. Gather necessary supplies					
2. Perform hand hygiene and put on PPE, if indicated					
3. Identify the patient.					
4. Assemble equipment on overbed table within reach.					
5. Close the curtains around the bed and close the door to the room, if possible. Explain to the patient what you are going to do					
6. Assess the patient for the possible need for analgesic medication . Administer appropriate prescribed analgesic. Allow enough time for the analgesic to achieve its effectiveness.					
7. Assist the patient to a comfortable position that provides easy access to the wound area.					
8. Check the position of drains, tubes, or other adjuncts before removing the dressing. Put on clean, disposable gloves and loosen tape on the old dressings					
9. Carefully remove the soiled dressings use small amounts of sterile saline to help loosen and remove it					
10. After removing the dressing, note the presence, amount, type, color, and odor of any drainage on the dressings Place soiled dressings in the appropriate waste receptacle. Remove your gloves and dispose of them in an appropriate waste receptacle					
11. Inspect the wound site for size, appearance, and drainage. Assess if any pain is present. Check the status of sutures, adhesive closure strips, staples, and drains or tubes, if present. Note any problems to include in your documentation.					
12. Using sterile technique, prepare a sterile work area and open the needed supplies					
13. Put on sterile gloves					
14. Clean the wound. Clean the wound from top to bottom and from the center to the outside.					
15. Once the wound is cleaned, dry the area using a gauze sponge in the same manner.					

16. Apply a layer of dry, sterile dressing over the wound. Forceps may be used to apply the dressing.					
17. Apply a surgical or abdominal pad (ABD) over the gauze at the site of the outermost layer of the dressing, as necessary					
18. Remove and discard gloves. Apply tape, Montgomery straps, or roller gauze to secure the dressings.					
19. After securing the dressing, label it with date and time. Remove all remaining equipment; place the patient in a comfortable position, with side rails up and bed in the lowest position.					
20. Remove PPE, if used. Perform hand hygiene.					

Performing irrigation of a wound

Objectives:

Students will be able to:

- Define a Wound Irrigation
 - Identify the goal of Wound Irrigation
 - Determine equipment's needed during the procedure
 - Performing irrigation of a wound
- Irrigation is a directed flow of solution over tissues. Wound irrigations are ordered to clean the area of pathogens and other debris and to promote wound healing. Irrigation procedures may also be
- ordered to apply heat or antiseptics locally, Normal saline is often the solution of choice when irrigating wounds.

➤ Equipment

- A sterile irrigation set, including a basin, irrigant container, and irrigation syringe
- Sterile irrigation solution as ordered, warmed to body temperature, commonly 0.9% normal saline solution
- Plastic bag or other waste container to dispose of soiled dressings
- Sterile gloves
- Sterile drape (may be optional)
- Clean, disposable gloves
- Moisture-proof gown, mask, and eye protection
- Additional PPE, as indicated
- Sterile dressing set or suture set (for the sterile scissors and forceps)
- Waterproof pad and bath blanket, as needed
- Sterile gauze dressings
- Sterile packing gauze, as needed
- Tape or ties
- Skin-protectant wipes

➤ Procedure

Steps	Rational
1. Review the medical order for wound care or the nursing plan of care related to wound care. Gather necessary supplies	Reviewing the order and plan of care validates the correct patient and correct procedure. Preparation promotes efficient time management and an organized approach to the task.
2. Perform hand hygiene and put on PPE, if indicated.	Hand hygiene and PPE prevent the spread of microorganisms. PPE is required based on transmission precautions.
3. Identify the patient	Identifying the patient ensures the right patient receives the intervention and helps prevent errors.
4. Close the curtains around the bed and close the door to the room if possible. Explain what you are going to do	This ensures the patient's privacy. Explanation relieves anxiety and facilitates cooperation.
5. Assess the patient for possible need for nonpharmacologic pain-reducing interventions or analgesic medication before wound care and/or dressing change. Administer appropriate prescribed analgesic. Allow enough time for the analgesic to achieve its effectiveness before beginning the procedure.	Pain is a subjective experience influenced by past experience. Wound care and dressing changes may cause pain for some patients.
6. Place a waste receptacle or bag at a convenient location for use during the procedure.	Having a waste container handy means the soiled dressing may be discarded easily, without the spread of microorganisms
7. Assist the patient to a comfortable position that provides easy access to the wound area. Position the patient so the irrigation solution will flow from the clean end of the wound toward the dirtier end. Use the bath blanket to cover any exposed area other than the wound. Place a waterproof pad under the wound site.	Patient positioning and use of a bath blanket provide for comfort and warmth. Gravity directs the flow of liquid from the least contaminated to the most contaminated area. Waterproof pad protects underlying surfaces
8. Carefully remove the soiled dressings. If any part of the dressing sticks to the underlying skin, use small amounts of sterile saline to help loosen and remove it	Cautious removal of the dressing is more comfortable for the patient and ensures that any drain present is not removed. Sterile saline moistens the dressing for easier removal and minimizes damage and pain.
9. After removing the dressing, note the presence, amount, type, color, and odor of	The presence of drainage should be documented. Proper disposal of soiled

any drainage on the dressings. Place soiled dressings in the appropriate waste receptacle.	dressings and used gloves prevents the spread of microorganisms.
10. Assess the wound for appearance, stage, presence of eschar, granulation tissue, necrosis, sinus tract, and drainage. Assess the appearance of the surrounding tissue	This information provides evidence about the wound healing process and/or the presence of infection.
11. Remove your gloves and put them in the receptacle	Discarding gloves prevents the spread of microorganisms
12. Set up a sterile field, if indicated, and wound cleaning supplies. Pour warmed sterile irrigating solution into the sterile container. Put on the sterile gloves	Using warmed solution prevents chilling the patient and may minimize patient discomfort. Sterile technique and gloves maintain surgical asepsis.
13. Position the sterile basin below the wound to collect the irrigation fluid	Patient and bed linens are protected from contaminated fluid.
14. Fill the irrigation syringe with solution.  Using your nondominant hand, gently apply pressure to the basin against the skin below the wound to form a seal with the skin 	The solution will collect in the basin and prevent the irrigant from running down the skin. Patient and bed linens are protected from contaminated fluid.

15.Gently direct a stream of solution into the wound. Keep the tip of the syringe at least 1 inch above the upper tip of the wound.



Debris and contaminated solution flow from the least contaminated to most contaminated area. High-pressure irrigation flow may cause patient discomfort as well as damage granulation tissue

16.Dry the surrounding skin with gauze dressings



Moisture provides a medium for growth of microorganisms. Excess moisture can contribute to skin irritation and breakdown.

17.Apply a skin protectant to the surrounding skin. Apply a new dressing to the wound



A skin protectant prevents skin irritation and breakdown. Dressings absorb drainage, protect the wound, and promote healing.

18.Remove and discard gloves. Apply tape, Montgomery straps, or roller gauze to secure the dressings.

Proper disposal of gloves prevents the spread of microorganisms. Tape or other securing products are easier to apply after gloves have been removed.

19.After securing the dressing, label it with date and time. Remove all remaining equipment; place the patient in a comfortable position, with side rails up and bed in the lowest position.	Recording date and time provides communication and demonstrates adherence to plan of care. Proper patient and bed positioning promotes safety and comfort.
20.Remove PPE, if used. Perform hand hygiene.	Removing PPE properly reduces the risk for infection transmission and contamination of other items. Hand hygiene prevents the spread of microorganisms

➤ Procedure Check List

PROCEDURE STEPS	Mark	Trial 1	Trial 2	Trial 3	comment
1. Review the medical order for wound care or the nursing plan of care related to wound care. Gather necessary supplies					
2. Perform hand hygiene and put on PPE, if indicated.					
3. Identify the patient					
4. Close the curtains around the bed and close the door to the room if possible. Explain what you are going to do					
5. Assess the patient for the possible need for analgesic medication. Administer appropriate prescribed analgesic. Allow enough time for the analgesic to achieve its effectiveness.					
6. Place a waste receptacle or bag at a convenient location for use during the procedure.					
7. Assist the patient to a comfortable position that provides easy access to the wound area.					
8. Carefully remove the soiled dressings use small amounts of sterile saline to help loosen and remove it					
9. After removing the dressing, note the presence, amount, type, color, and odor of any drainage on the dressings. Place soiled dressings in the appropriate waste receptacle.					
10. Assess the wound for appearance, stage, presence of eschar, granulation tissue, necrosis, sinus tract, and drainage. Assess the appearance of the surrounding tissue					
11. Remove your gloves and put them in the receptacle					
12. Set up a sterile field, if indicated, and wound cleaning supplies. Pour warmed sterile irrigating solution into the sterile container. Put on the sterile gloves					
13. Position the sterile basin below the wound to collect the irrigation fluid					
14. Fill the irrigation syringe with solution. Using your nondominant hand, gently apply pressure to the basin against the skin below the wound to form a seal with the skin					
15. Gently direct a stream of solution into the wound. Keep the tip of the syringe at least 1 inch above the upper tip of the wound.					

16. Dry the surrounding skin with gauze dressings					
17. Apply a skin protectant to the surrounding skin. Apply a new dressing to the wound					
18. Remove and discard gloves. Apply tape, Montgomery straps, or roller gauze to secure the dressings.					
19. After securing the dressing, label it with date and time. Remove all remaining equipment; place the patient in a comfortable position, with side rails up and bed in the lowest position.					
20. Remove PPE, if used. Perform hand hygiene.					

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Chest Physiotherapy (CPT)

Objectives:

Students will be able to:

- Define Chest Physiotherapy (CPT).
- Identify the goal of Chest Physiotherapy (CPT).
- Determine Positions of Chest Physiotherapy (CPT).
- Determine equipment's needed during the procedure.
- Applying Chest Physiotherapy (CPT).

Preparing Client for CPT

Equipment

- Hospital bed or other surface to place client in head down position
- Gown or towel (optional)
- Tissues
- Container for sputum
- Clean gloves
- Stethoscope
- Pulse oximeter (if indicated)
- Mouthwash/oral hygiene product

Procedure	
Action	Rational
1. Validate physician's order for CPT.	These procedures are tiring to clients, time-consuming, and contraindicated in cases of osteoporosis, pulmonary embolism, cardiac conditions, lung cancer, or other conditions not associated with excessive sputum production.
2. Perform hand hygiene.	
3. Administer CPT before or at least 2 hr after meals to prevent vomiting	
4. Establish the location of lung segments if the entire lung field is to undergo CPT; the affected segment should be drained first.	Usually, the lower areas are the most affected
5. Provide privacy.	
6. Prepare client by explaining CPT and purpose.	
7. Auscultate chest for breath sounds and adventitious sounds before therapy.	

8 .Obtain pulse oximetry (SPO2) if indicated before therapy	
9. Place towel over skin when performing CPT (optional).	

Performing Postural Drainage

Procedure	
Action	Rational
1 .Loosen any tight clothing. 2 .Lower head of bed slowly so that client's head is positioned at no greater than a 25° downward angle. 3 .Place sputum container and tissues in client's reach. 4 .Tell client to remain in position for 3–15 minutes. 5.Instruct client to expectorate secretions. 6 .Instruct client to turn to other side, then to supine position, then repeat procedure. NOTE: Clients should deep breathe between position changes. 7.Assist client to slowly return to normal sitting position after coughing in dependent positions. 8.Determine pulse oximetry if ordered. 9 .Auscultate chest areas for improved breath sounds. 10 .Note character and measure sputum, then discard. 11 .Remove gloves and perform hand hygiene. 12. Offer oral hygiene after secretion expectoration.	Gravity helps mobilize secretions Percussion and vibration may be added to help mobilize secretions (skills to follow).

Performing Chest Percussion

Action	Rational
1.Cover area to be percussed with gown or cloth towel (optional). 2 .Holding arms with elbows slightly flexed, cup your hands with thumbs and fingers closed. Keeping wrists loose and relaxed, rhythmically flex and extend wrists to clap over area to be drained.	This motion produces vibrations that loosen secretions for easier removal with coughing or suctioning.

- 3 .Percuss by alternating hands and listen for hollow sound with strikes.
- 4 .Slowly and rhythmically percuss each area for 3-5 minutes.
- 5 .Do not percuss over bony prominences, breasts, or tender areas.
- 6 .Encourage client to “huff” cough after percussion of lung areas.
- 7 .Auscultate all lung areas for changes in breath sounds.
- 8 .Don gloves, note character and measure quantity of sputum, and discard.
- 9 .Remove gloves and perform hand hygiene.
- 10 .Offer oral hygiene.
11. Document procedure and client’s response.

Vibrations are not transmitted to the chest wall through bone or breast tissue and percussion in these areas can cause discomfort.



Perform percussion and vibration with each position change.

Performing Chest Vibration

Action

- 1 .Perform vibration after postural drainage and percussion in each position.
- 2 .Cover area to be vibrated with gown or towel (optional).
3. Instruct client to breathe in through nose and exhale slowly through pursed lips.
- 4 .Place your hands flat over area to be vibrated or place one hand on top of the other. Keep your arms and shoulders straight and wrists stiff.
5. Have client inhale deeply.
- 6.As client exhales through pursed lips, use moderate pressure to vibrate chest by quickly contracting and relaxing your arms and shoulders.
- 7 .Vibrate for three or four exhalations over the area.
- 8 .Encourage clients to “huff” cough before changing positions.
- 9 .Assess vital signs, pulse oximetry, and auscultate breath sounds.
- 10 .Don gloves.

Rational



Place hands flat over area to be vibrated, keeping wrists stiff.

This increases the turbulence and velocity of exhaled air and loosens and mobilizes secretions from peripheral airways.

11.Measure and note character of expectorated secretions, then discard. 12.Remove gloves and perform hand hygiene. 13 .Provide oral hygiene. 14.Document procedure and client’s response.	
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Using an Airway Clearance System (Wrap)

Equipment

See Preparing Client for CPT

Disposable inflatable chest wraps of appropriate size (small, medium, large, or extra-large)

Air pulse generator with connecting tubes

Preparation

- 1 .Validate physician’s order for high-frequency chest wall oscillation therapy, frequency, and duration of therapy.
2. Perform hand hygiene.
- 3 .Identify client by name and birth date.
4. Provide privacy.
5. Explain function and purpose of vest/wrap.
6. Assess lung sounds before therapy.

Action	Rational
1 .Elevate head of bed or have client sit in chair. Alternately, place wrap on bed and roll client onto wrap. 2 .Apply wrap over single layer of clothing, wrapping deflated vest around client’s chest, just beneath axillae; have client inhale deeply; and secure with Velcro fasteners. 3.Adjust generator to a comfortable working height. 4 .Lock generator castors. _____ Rationale: To prevent movement. 5 .Slide air hose into connector port on front of generator using a twist and push motion. 6 .Slide other end of hose into disposable vest slits. 7 .Fasten Velcro loops to secure air hoses in position. 8 .Plug the power cord into the power inlet on back of air pulse generator. 9 .Plug power cord into grounded outlet.	Wrap should fit snugly, but not be uncomfortable.

- 10 .Adjust arrows up or down to set frequency, pressure, and treatment time, confirming that setting on screen matches those prescribed.
- 11 .Press “ON” button to inflate disposable vest.
- 12 .Press “OFF” button to end treatment before set time and wrap will deflate.
- 13“ .Session complete” message will show on screen when treatment set time is complete and wrap will deflate.
- 14 .Unplug system and remove disposable air hoses and vest from client.
- 15 .Store vest and air hoses in client’s area and return generator to storage for future use.
- 16 .Reassess client’s breath sounds; note character and amount of coughed/suctioned secretions.
- 17 .Assist client to a comfortable position.
18. Perform hand hygiene.



Vest/wrap high-frequency chest wall oscillation therapy increases mobilization and clearance of airway secretions.

Air pulsation provides vibration to client’s chest. Treatment will last 10 minutes, but then must be restarted if indicated. Usual treatments are 10–20 minutes.

Maintaining Chest Gravity Drainage/Suction Bottle System

Students will be able to:

- Define Chest tube
- Identify the goal of Chest tube
- Determine equipment's needed during the procedure.
- Applying care of Chest tube.

- Three-bottle chest drainage system in holder with
- manufacturer's instructions for use Tape
- Suction source, if indicated (e.g., wall suction or Emerson pump)
- Clean gloves and personal protective equipment

the procedure.

older with
(suction or Emerson pump)
equipment

Rational

The bottle stopper has two openings, one for inlet from the client, one for outlet to the atmosphere. Air vent allows air from client's intra pleural space to escape and to maintain negative pressure relative to client's intra pleural pressure.

Dependent loops will inhibit air removal.

9 .Check tubing for obstruction.	
10 .Report event of continuous bubbling in the water seal bottle.	This is sign that air is being evacuated from the client's intra pleural space
11 .Assess for return of breath sounds or obtain chest x-ray to determine re expansion.	Tidaling ceases when lung has re-expanded or tube system is obstructed.
	This is indicative of an air leak.

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Applying A Cardiac Monitor

Objectives:

Students will be able to:

- Define cardiac monitoring.
- Identify the goal of Cardiac monitoring.
- Determine Positions of cardiac monitor electrode.
- Determine equipment's needed during the procedure.
- Applying A Cardiac Monitor.

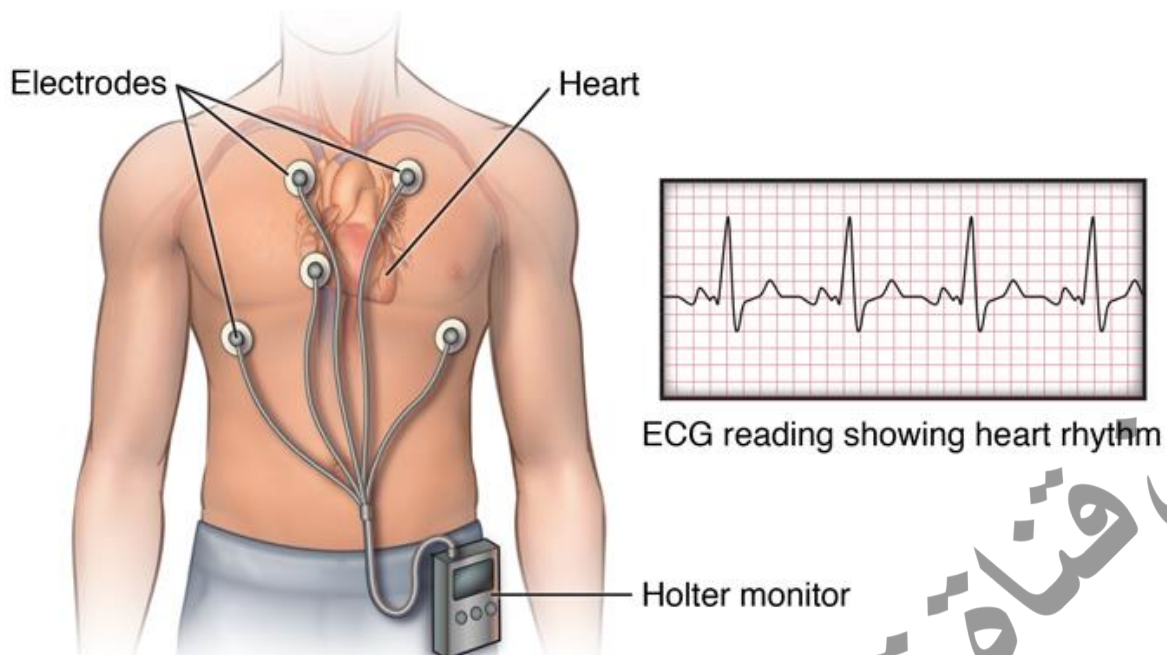
➤ Definition of cardiac monitoring

- Bedside cardiac monitor is a device “hard-wired monitor or a wireless” used to provide continuous observation of heart’s electrical activity. (Larson & Brady, 2008).

➤ Goal of Cardiac Monitoring

- Cardiac monitoring is used for patients with conduction disturbances and for those at risk for life threatening arrhythmias, such as postoperative patients and patients who are sedated.
- Cardiac monitoring is obtained using three or five electrodes and leads on the patient the patient’s chest.
- Telemetry uses a small transmitter connected to an ambulatory patient to send electrical signals to another location, where they are displayed on a monitor screen. Telemetry is especially useful for monitoring arrhythmias that occur during sleep, rest, exercise, or stressful situations.





➤ When to use

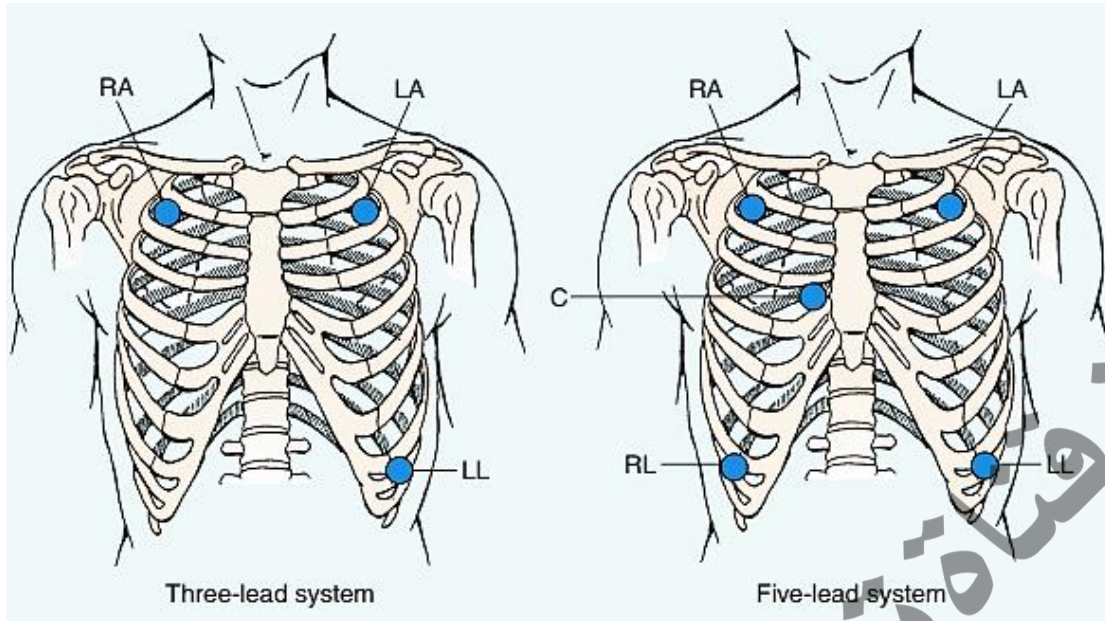
- 1- Patient in ICU and CCU.
- 2- During operation and postoperative period.
- 3- Patient with cardiac dysrhythmia.

➤ Equipment

- 1- Lead wires.
- 2- Bedside cardiac monitor or telemetry transmitter.
- 3- Pregel led (gel foam) electrodes (number varies from 3 to 5).
- 4- patient cable for hardwire cardiac monitoring.
- 5- Transmitter, transmitter pouch, and telemetry battery pack for telemetry
- 6- Skin cleanser.
- 7- Clean, dry towel, washcloth, or gauze.
- 8- Hair clippers (optional depending on hair at electrode sites).
- 9- PPE, as indicated.



➤ Positions of cardiac monitor electrode



✓ Positions for the three-lead system:

- 1- RA (white electrode) below right clavicle, second ICS, right midclavicular line.
- 2- LA (black electrode) below left clavicle, second ICS, left midclavicular line.
- 3- LL (red electrode) left lower ribcage, eighth ICS, left midclavicular line.

✓ Positions for five-lead system:

- 1- RA (white electrode) below right clavicle, second ICS, right midclavicular line.
- 2- RL (green electrode) right lower ribcage, eighth ICS, right midclavicular line.
- 3- LA (black electrode) below left clavicle, second ICS, left midclavicular line.
- 4- LL (red electrode) left lower ribcage, eighth ICS, left midclavicular line.
- 5- Chest (brown electrode) any V lead position, usually V1 (fourth ICS, right sternal border).

➤ Procedure

Steps	Rational
1. Verify the order for cardiac monitoring on the patient's medical record. Identify the patient.	This ensures that the correct intervention is performed on the correct patient. Identifying the patient ensures the right patient receives the intervention and helps prevent errors.
2. Gather equipment.	Provides for an organized approach to the task.
3. Perform hand hygiene and put on PPE, if indicated.	Hand hygiene and PPE prevent the spread of microorganisms.
4. Close curtains around the bed and close the door to the room, if possible. Explain the procedure to the patient.	To keep privacy and decrease patient concerns

5. Check skin for excess oil or moisture. If present, wipe chest or limbs with clean, dry towel. Clean and prepare chest area for electrode placement with soap and water.	Provides clear, accurate recording without artifact.
6. Prepare patient for procedure: ✓ Remove or reposition patient's gown to expose only patient's chest. Keep abdomen and thighs covered. ✓ Place patient in supine position.	Facilitates correct placement of cardiac leads and maintains patient's modesty.
7. Apply electrodes in correct positions for either a three or five-electrode system. 	Proper placement of leads is important for accurate dysrhythmia interpretation.
8. Attach monitor leads to electrodes. Colors of leads represent their polarity.	Coloring system allows for consistent application of leads.
9. Check bedside monitor or telemetry station for any messages indicating electrode or lead issues. Troubleshoot as needed.	Monitoring system itself may detect bad electrode contact with skin or loose connection.
10. Set the upper and lower limits of the heart rate alarm, based on the patient's condition or unit policy.	Setting the alarm allows for audible notification if the heart rate is beyond limits.
11. Return the patient to a comfortable position. Lower the bed height and adjust the head of bed to a comfortable position.	To maintain the patient comfort.
12. Remove additional PPE, if used. Perform hand hygiene.	To prevent transmission of microorganism.

➤ Nursing consideration

- 1- The application of a cardiac monitor is not delegated to nursing assistive personnel.
- 2- Review the patient's medical record and plan of care for information about the patient's need for cardiac monitoring.
- 3- Assess the patient's cardiac status, including heart rate, blood pressure, and auscultation of heart sounds.

- 4- Inspect the patient's chest for areas of irritation, breakdown, or excessive hair that might interfere with electrode placement "Electrode sites must be dry, with minimal hair".
- 5- The patient may be sitting or supine, in a bed or chair.
- 6- Review alarm trends and waveforms at least once per shift and on report of an alarm.
- 7- Record at least one rhythm strip per shift per agency policy, either on paper or save to electronic health record (EHR).
- 8- Report any unexpected outcomes immediately to the health care provider.

➤ Procedure Check List

Procedure Steps	Mark	Trial 1	Trial 2	Trial 3	comment
1. Verify the order for cardiac monitoring on the patient's medical record. Identify the patient.					
2. Gather equipment.					
3. Perform hand hygiene and put on PPE, if indicated.					
4. Close curtains around the bed and close the door to the room, if possible. Explain the procedure to the patient.					
5. Check skin for excess oil or moisture. If present, wipe chest or limbs with clean, dry towel. Clean and prepare chest area for electrode placement with soap and water.					
6. Prepare patient for procedure: ✓ Remove or reposition patient's gown to expose only patient's chest. Keep abdomen and thighs covered. ✓ Place patient in supine position.					
7. Apply electrodes in correct positions for either a three or five-electrode system.					
8. Attach monitor leads to electrodes. Colors of leads represent their polarity.					
9. Check bedside monitor or telemetry station for any messages indicating electrode or lead issues. Troubleshoot as needed.					
10. Set the upper and lower limits of the heart rate alarm, based on the patient's condition or unit policy.					
11. Return the patient to a comfortable position. Lower the bed height and adjust the head of bed to a comfortable position.					
12. Remove additional PPE, if used. Perform hand hygiene.					

Using A Pulse Oximeter

Objectives:

Students will be able to:

- Define Pulse Oximeter.
- Identify the goal of using Pulse Oximeter.
- Enumerate Conditions That May Produce False Readings of Pulse Oximeter.
- Determine equipment's needed during the procedure.
- using A Pulse Oximeter.

➤ Definition

- Pulse oximetry is the noninvasive measurement of arterial blood oxygen saturation, the percent to which hemoglobin is filled with oxygen. A pulse oximeter is a probe with a light-emitting diode (LED) connected by cable to an oximeter.

➤ the goal of using Pulse Oximeter

- Pulse oximetry is useful for monitoring patients receiving oxygen therapy, monitoring those at risk for hypoxia, monitoring those at risk of hypoventilation (opioids use, neurologic compromise), and postoperative patients.
- Desaturation (decreased level of SpO₂) indicates gas exchange abnormalities. And a late sign of respiratory compromise in patients with reduced rate and depth of breathing.
- Pulse oximetry does not replace arterial blood gas analysis. Sensors are available for use on a finger, a toe, a foot (infants), an earlobe, forehead, and the bridge of the nose. It is important to use the appropriate sensor for the intended site.

FINGER PULSE OXIMETER



HANDHELD PULSE OXIMETER



TABLETOP PULSE OXIMETER



WRIST-WORN PULSE OXIMETER



PEDIATRIC PULSE OXIMETERS



➤ Equipment

- 1- Oximeter.
- 2- Oximeter probe appropriate for patient and recommended by oximeter manufacturer.
- 3- Acetone or nail polish remover if needed.
- 4- Pen and vital sign flow sheet or electronic health record (EHR).

Spo2	Interventions
≥95%	• Considered normal and generally requires no invasive intervention. • Continue routine monitoring of patient.
91%–94%	• Considered borderline. • Assess and adjust probe placement. • Begin oxygen at 2 L/min titrated to SpO₂ >95%.
85%–90%	• Elevate head and encourage patient to cough and breathe deeply. • Assess airway and suction as needed. • Administer oxygen and titrate to SpO₂ >95%. • If condition fails to improve, assist ventilations manually and prepare to intubate.
<85%	• Administer 100% oxygen, sit patient upright, encourage coughing and deep breathing, and suction as needed. • Assist ventilations manually and prepare to intubate if condition fails to improve. • Consider reversal agents for possible drug induced respiratory depression.

➤ Conditions That May Produce False Readings

✓ False High Readings

- 1- Anemia.
- 2- Alkalosis.
- 3- CO poisoning.
- 4- Hypovolemia.
- 5- patient movement.

✓ False Low Readings

- 1- Cool extremities.
- 2- Drugs (vasoconstrictors).
- 3- Nail polish/nail infection.
- 4- patient movement.
- 5- Poor peripheral circulation.
- 6- Reynaud's disease.

➤ Procedure

Steps	Rational
1. Review health record for any health problems that would affect the patient's oxygenation status.	Identifying influencing factors aids in interpretation of results.
2. Bring necessary equipment.	Bringing everything to the bedside conserves time and energy.
3. Perform hand hygiene and put on PPE, if indicated.	Hand hygiene and PPE prevent the spread of microorganisms.
4. Identify the patient.	To ensure that the right patient receive the right intervention.
5. Close the curtains around the bed and close the door to the room, if possible. Explain the procedure.	To keep privacy and reduce patient concern.
6. Assess for signs and symptoms of alterations in oxygen saturation (e.g., altered respiratory rate, depth, or rhythm; adventitious breath sounds; cyanotic nails, lips, mucous membranes, or skin; restlessness; difficulty breathing).	Physical signs and symptoms indicate abnormal oxygen saturation.
7. Assess for factors that influence measurement of SpO ₂ (e.g., oxygen therapy, respiratory therapy such as postural drainage and percussion, hemoglobin level, hypotension, temperature, nail polish, and medications such as bronchodilators).	Allows nurse to assess oxygen saturation variations accurately.
8. Determine most appropriate patient-specific site (e.g., finger, earlobe, bridge of nose, forehead) for sensor probe placement by measuring capillary refill time. If capillary refill time is less than 2 seconds, select alternative site.	Changes in SpO ₂ level are reflected in the circulation of finger capillary beds within 30 seconds and in the capillary beds of earlobes within 5 to 10 seconds.



9. Attach sensor to monitoring site. If using finger, remove fingernail polish from digit with acetone or polish remover.



Select sensor site based on peripheral circulation and extremity temperature.

10. Once sensor is in place, turn on oximeter by activating power.



11. Leave sensor in place 10 to 30 seconds or until oximeter readout reaches constant value and pulse display reaches full strength.



Reading usually takes 10 to 30 seconds, depending on site selected.

12. If you plan to monitor oxygen saturation continuously, verify SpO₂ alarm limits preset by the manufacturer at a low of 85% and a high of 100%.

Alarms must be set at appropriate limits and volumes to avoid frightening patients and visitors.

13. Compare SpO₂ reading with patient's previous baseline and acceptable SpO₂ values.

Allows nurse to assess for change in patient's condition and presence of respiratory alteration.

14. Remove PPE, if used. Perform hand hygiene.

reduces the spread of microorganisms.

➤ Procedure checklist

Procedure Steps	Mark	Trial 1	Trial 2	Trial 3	comment
1. Review health record for any health problems that would affect the patient's oxygenation status.					
2. Bring necessary equipment.					
3. Perform hand hygiene and put on PPE, if indicated.					
4. Identify the patient.					
5. Close the curtains around the bed and close the door to the room, if possible. Explain the procedure.					
6. Assess for signs and symptoms of alterations in oxygen saturation (e.g., altered respiratory rate, depth, or rhythm; adventitious breath sounds; cyanotic nails, lips, mucous membranes, or skin; restlessness; difficulty breathing).					
7. Assess for factors that influence measurement of SpO ₂ (e.g., oxygen therapy, respiratory therapy such as postural drainage and percussion, hemoglobin level, hypotension, temperature, nail polish, and medications such as bronchodilators).					
8. Determine most appropriate patient-specific site for sensor probe placement by measuring capillary refill time. If capillary refill time is less than 2 seconds, select alternative site.					
9. Attach sensor to monitoring site. If using finger, remove fingernail polish.					
10. Once sensor is in place, turn on oximeter by activating power.					
11. Leave sensor in place 10 to 30 seconds or until oximeter readout reaches constant value and pulse display reaches full strength.					
12. verify SpO ₂ alarm limits preset by the manufacturer at a low of 85% and a high of 100%.					
13. Compare SpO ₂ reading with patient's previous baseline and acceptable SpO ₂ values.					
14. Remove PPE, if used. Perform hand hygiene.					

➤ **Nursing consideration**

- 1- The measurement of oxygen saturation using a pulse oximeter may be delegated to nursing assistive personnel.
- 2- Assess the patient's skin temperature and color, including the color of the nail beds. Temperature is a good indicator of blood flow.
- 3- Auscultate the lungs. Note the amount of oxygen and delivery method if the patient is receiving supplemental oxygen.
- 4- Instructing to notify nurse immediately of any reading lower than SpO₂ of 95% or value for specific patient.
- 5- Instructing to prevent to use pulse oximetry to obtain heart rate because oximeter will not detect an irregular pulse.

Obtaining an Electrocardiogram (ECG)

Objectives:

Students will be able to:

- Define Pulse Electrocardiogram (ECG).
- Identify The goal of Obtaining an Electrocardiogram (ECG).
- Enumerate The conduction pathway.
- Determine equipment's needed during the procedure.
- Determine the Placement of chest and limb leads
- Demonstrate an Electrocardiogram (ECG) procedure

➤ Definition

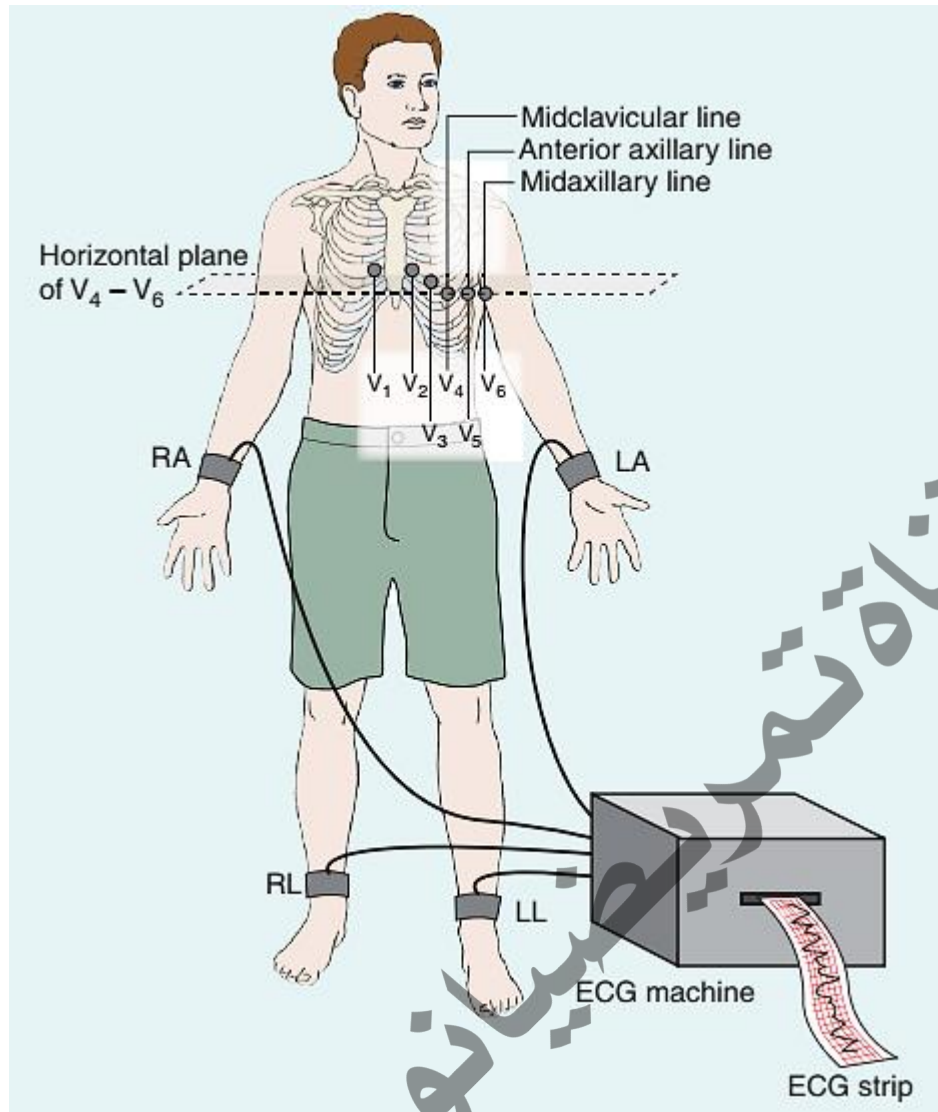
- Electrocardiography “ECG” [also abbreviated as EKG in some references] is one of the most valuable and frequently used diagnostic tools. ECG measures the heart’s electrical activity.
- Impulses moving through the heart’s conduction system create electric currents that can be monitored on the body’s surface. Electrodes attached to the skin can detect these electric currents and transmit them to an instrument that produces a record, the electrocardiogram, of cardiac activity. The data are graphed as waveforms.



Electrocardiogram (waveform) strip

➤ The goal of Obtaining an Electrocardiogram (ECG).

- ECG can be used to identify myocardial ischemia and infarction, rhythm and conduction disturbances, chamber enlargement, electrolyte imbalances, and drug toxicity.
- The standard 12-lead ECG uses a series of electrodes placed on the extremities and the chest wall to assess the heart from 12 different viewpoints (leads) by attaching ten cables with electrodes to the patient’s limbs and chest: four limb electrodes and six chest electrodes

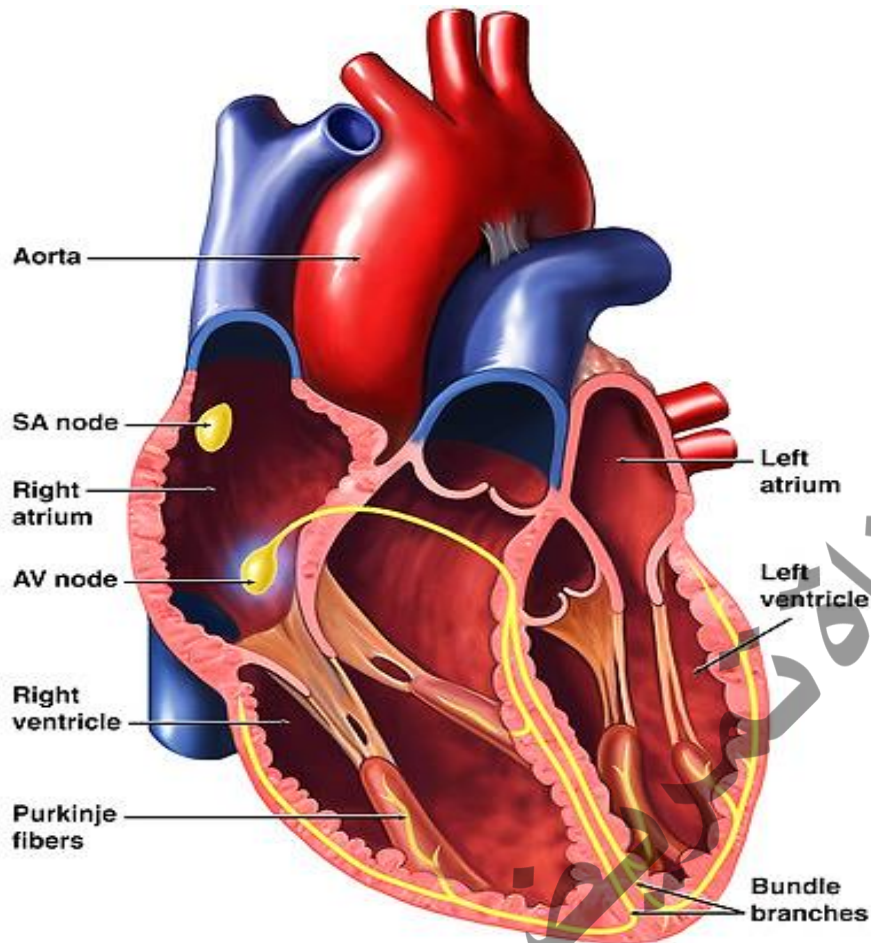


➤ **Conductive system of the heart**

- Cardiac conduction is the pathway at which the heart conducts electrical impulses. These impulses cause the heart to contract and then relax. The constant cycle of heart muscle contraction followed by relaxation cause blood to be pumped throughout the body.

✓ **The conduction pathway is made up of five elements:**

- 1 - the Sino-atriale (SA) node,
- 2 - the atrioventricular (AV) node,
- 3 - the bundle of His,
- 4 - the left and right bundle branches,
- 5 - the Purkinje fibers.



➤ Equipment

- 1- ECG machine.
- 2- Recording paper.
- 3- Disposable pregelled electrodes.
- 4- Adhesive remover swabs.
- 5- 4 × 4 gauze pads.
- 6- Skin cleanser and water, if necessary.
- 7- Additional PPE, as indicated.
- 8- Hair clippers (optional depending on hair at electrode sites).
- 9- Bath blanket.

➤ Placement of chest and limb leads

Chest leads

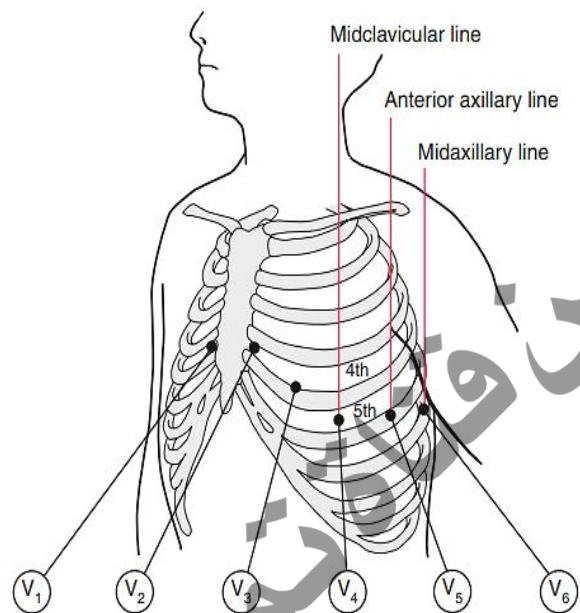
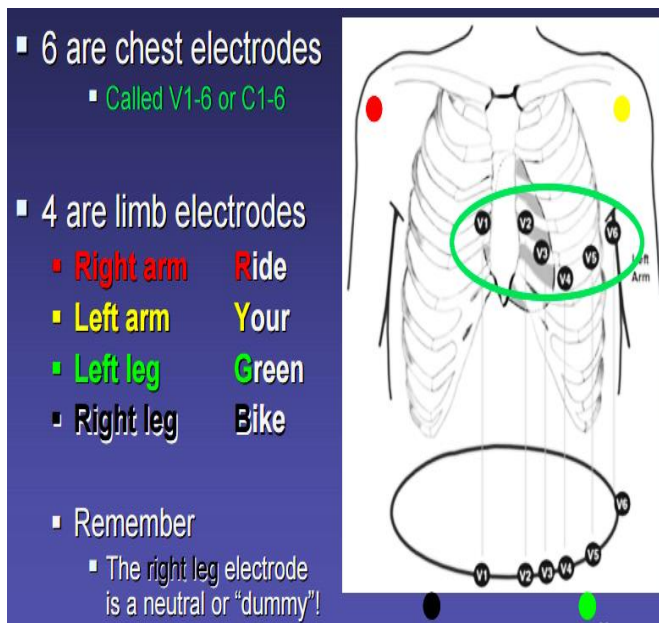
- V1 – 4th intercostal space (ICS) at right sternal angle.
- V2 – 4th ICS at left sternal border.
- V3 – Midway between V2 and V4.
- V4 – 5th ICS at midclavicular line.
- V5 – Left anterior axillary line at level of V4 horizontally.
- V6 – Left midaxillary line at level of V4 horizontally.

Limb leads

- RA = Right Arm.
- LA = Left Arm.

RL = Right Leg.

LL = Left Leg.



➤ Procedure

Steps	Rational
1. Verify the order for an ECG in the patient's medical record.	This ensures that the correct intervention is performed on the correct patient.
2. Gather all equipment.	Assembling equipment provides for an organized approach to the task.
3. Perform hand hygiene and put on PPE, if indicated.	Hand hygiene and PPE prevent the spread of microorganisms.
4. Identify the patient.	Identifying the patient ensures the right patient receives the intervention and helps prevent errors.
5. Close curtains around the bed and close the door to the room, if possible. As you set up the machine to record a 12-lead ECG, explain the procedure to the patient.	To keep privacy and reduce patient concern
6. Place the ECG machine close to the patient's bed, and plug the power cord into the wall outlet.	Having equipment available saves time and facilitates accomplishment of the task.
7. Have the patient lie supine in the center of the bed with the arms at the sides. Raise the head of the bed if necessary to promote comfort. Expose the patient's arms and legs, and drape appropriately. Encourage the patient to relax the arms and legs. Ensure the wrists do not	Proper positioning helps increase patient comfort and will produce a better tracing. Proper positioning and relaxation of the arms and legs minimizes muscle tension and

touch the waist. Make sure the feet do not touch the bed's footboard.

trembling and electrical interference.

8. Clean and prepare isolated electrode area with soap and water. Wipe area with rough washcloth or gauze. If an area is excessively hairy, clip the hair. Do not shave hair.

Proper skin preparation before ECG electrodes are placed decreases skin impedance and signal noise, Shaving causes micro abrasions on the chest skin.

9. Apply electrodes in correct positions. Placed directly on the chest wall and not the breast tissue. The breast may need, to be lifted to accommodate chest leads (V4, V5, V6.)

Proper placement of leads is important for accurate interpretation of 12-lead ECG.



10. After the application of all the leads, ensure that the cables are not pulling on the electrodes or lying over each other. Make sure the paper speed selector is set to the standard **25 m/second** and that the machine is set to full voltage, **10 mv**.



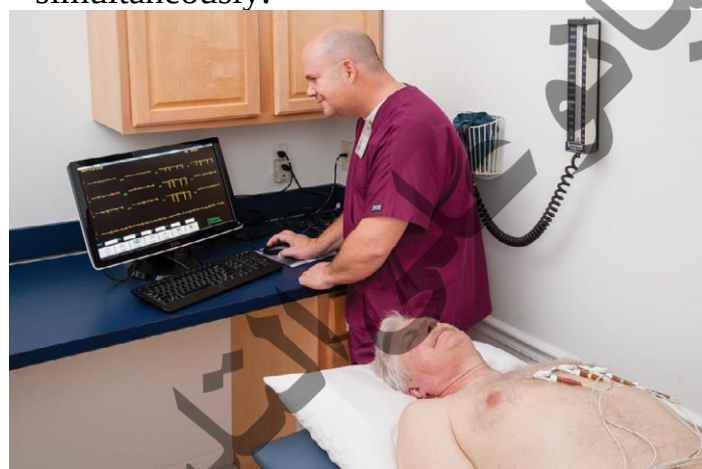
Minimizes electrical artifact and improves quality and accuracy of the ECG

11. If necessary, enter the appropriate patient identification data into the machine.

This allows for proper identification of the ECG strip.

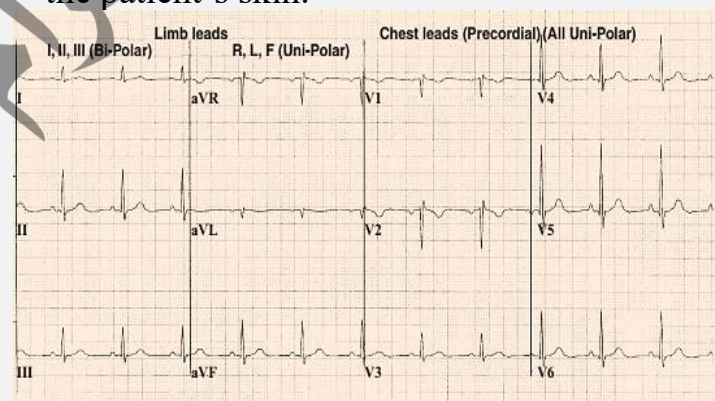
12. Press the AUTO button. Observe the tracing quality. The machine will record all 12 leads automatically, recording 3 consecutive leads simultaneously.

Observation of tracing quality allows for adjustments to be made, if necessary.



13. When the machine finishes recording the 12-lead ECG, remove the electrodes and clean the patient's skin.

Removal and cleaning promote patient comfort.



14. After disconnecting the lead wires from the electrodes, dispose of the electrodes. Return the patient to a comfortable position.	Proper disposal deters the spread of microorganisms. Positioning with head adjustment promotes patient comfort.
15. Clean the ECG machine.	Cleaning equipment between patient uses decreases the risk for transmission of microorganisms.
16. Remove additional PPE, if used. Perform hand hygiene.	Removing PPE properly reduces the risk for infection transmission and contamination of other items.

➤ Procedure checklist

Procedure Steps	Mark	Trial 1	Trial 2	Trial 3	comment
1. Verify the order for an ECG in the patient's medical record.					
2. Gather all equipment.					
3. Perform hand hygiene and put on PPE, if indicated.					
4. Identify the patient.					
5. Close curtains around the bed and close the door to the room, if possible. As you set up the machine to record a 12-lead ECG, explain the procedure to the patient.					
6. Place the ECG machine close to the patient's bed, and plug the power cord into the wall outlet.					
7. Have the patient lie supine in the center of the bed with the arms at the sides. Raise the head of the bed if necessary to promote comfort. Expose the patient's arms and legs, and drape appropriately.					
8. Clean and prepare isolated electrode area with soap and water. Do not shave hair.					
9. Apply electrodes in correct positions. Placed directly on the chest wall and not the breast tissue. The breast may need, to be lifted to accommodate chest leads (V4, V5, V6.) • V1—4th intercostal space (ICS) at right sternal border. Females: Choose a site as close to standard position as possible. • V2—4th ICS at left sternal border. • V3—Midway between V2 and V4. • V4—5th ICS at midclavicular line.					

• V5—Left anterior axillary line at level of V4 horizontally. • V6—Left midaxillary line at level of V4 horizontally					
10. ensure that the cables are not pulling on the electrodes or lying over each other. Make sure the paper speed selector is set to the standard 25 m/second and that the machine is set to full voltage, 10 mv .					
11. If necessary, enter the appropriate patient identification data into the machine.					
12. Press the AUTO button. Observe the tracing quality. The machine will record all 12 leads automatically, recording 3 consecutive leads simultaneously.					
13. When the machine finishes recording the 12-lead ECG, remove the electrodes and clean the patient's skin.					
14. After disconnecting the lead wires from the electrodes, dispose of the electrodes. Return the patient to a comfortable position.					
15. Clean the ECG machine.					
16. Remove additional PPE, if used. Perform hand hygiene.					

➤ Nursing consideration

- 1- Obtaining an electrocardiogram is not delegated to nursing assistive personnel.
- 2- Review the patient's medical record and plan of care for information about the patient's need for an ECG.
- 3- Assess the patient's cardiac status, including heart rate, blood pressure, and auscultation of heart sounds.
- 4- If the patient is already connected to a cardiac monitor, remove the electrodes to accommodate the precordial leads and minimize electrical interference on the ECG tracing.
- 5- Keep the patient away from objects that might cause electrical interference, such as equipment, fixtures, and power cords.
- 6- Inspect the patient's chest for areas of irritation, breakdown, or excessive hair that might interfere with electrode placement.

Nasogastric tube feeding

Objectives:

Students will be able to:

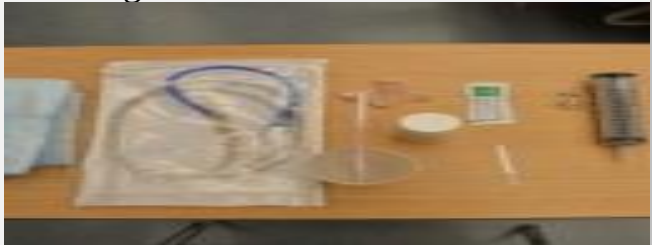
- Define Nasogastric tube feeding.
- Identify the goal of Nasogastric tube feeding.
- Determine equipment's needed during the procedure.
- Applying Nasogastric tube feeding.

Definition: Nasogastric (NG) feeding is where a narrow feeding tube is placed through your nose down into your stomach. The tube can be used to give you fluids, medications and liquid food complete with nutrients directly into your stomach.

➤ Steps for inserting a nasogastric tube

Safety considerations:

- Perform hand hygiene.
- Introduce yourself to patient.
- Confirm patient ID using two patient identifiers (e.g., name and date of birth).
- Explain process to patient.
- Listen and attend to patient cues.
- Ensure patient's privacy and dignity.

Steps	Additional information
1.Perform hand hygiene and gather supplies.	This prevents the transmission of microorganisms 
2. Visually inspect condition of patient's nasal and oral cavities.	Check for signs of infection or skin breakdown.
3. Assess for the best nostril before you begin. Do this by occluding one side and asking the patient to sniff. Ask the patient about previous injuries or history of a deviated septum.	If either nostril is equally suitable, select the nostril closest to the suction. 
4. Palpate patient's abdomen for distension, pain, and/or rigidity. Auscultate for bowel sounds.	Document assessment findings and determine appropriateness of NG tube insertion related to reason for insertion and patient's physical assessment.
5. Assess patient's level of consciousness and understanding of procedure.	Patient must be able to follow instructions related to NG insertion to allow for

	passage of tube through nasal and gastrointestinal tracts.
6. Check doctor's orders for type of NG tube to be placed and reason for placement.	Check appropriate orders relevant to patient safety.
7. Check doctor's orders to determine whether the NG tube is to be attached to suction or a drainage bag.	This should be commensurate with the reason for the NG tube.
8. Position patient sitting up at 45 to 90 degrees (unless contraindicated by the patient's condition), with a pillow under the head and shoulders.	This allows the NG tube to pass more easily through the nasopharynx and into the stomach.
9. Raise bed to a comfortable working height.	This helps prevent biomechanical injury to the health care provider.
10. Place a towel on the patient's chest and provide facial tissues and an emesis basin.	Nasal and oral secretions may be evident during the procedure.
11. Provide patient with drinking water and a straw if the patient is not fluid restricted.	Sipping water through a straw helps to initiate the swallowing reflex and facilitate passing of NG tube
12. Stand on patient's right side if you are right-handed and the left side if you are left-handed.	You will use your dominant hand to insert the tube.
13. Measure distance of the tube from The tip of the nose, to... The earlobe, to... The xiphoid process and then mark the tube at this point.	 Measure from tip of nose to earlobe  Measure from earlobe to xiphoid
14. Lubricate NG tube tip	

15. Curve 10 to 15 cm of the end of the NG tube around your gloved finger, and then release it.	 Curling the NG tube around your finger helps it conform to the normal curve of the nasopharynx.
16. Have patient drop head forward and breathe through the mouth.	Dropping the head forward closes the trachea and opens the esophagus, which allows the NG tube to pass more easily through the nasopharynx and into the stomach
17. Insert NG tube tip slowly into the patient's nostril and advance it steadily, in a downward direction, along the bottom of the nasal passage, with the curved end pointing downward in the direction of the ear on the same side as the nostril.	
18. You may feel slight resistance as you advance along the nasal passage. Twist the tube slightly, apply downward pressure, and continue trying to advance the tube. If significant resistance is felt, remove the tube and allow the patient to rest before trying again in the other nostril.	 Advance the tube gently
19. If there is difficulty in passing the NG tube, you may ask the patient to sip water slowly through a straw unless oral fluids are contraindicated. If oral fluids are not allowed, ask the patient to try dry swallowing while you advance the tube.	If patient continues to gag or cough, check that the tube is not coiled in the back of the mouth, using a tongue blade and a flashlight to check the back of the mouth.
20. Continue to advance NG tube until you reach the mark/tape you had placed for measurement.	This ensures accurate placement.
21. Temporarily anchor the tube to patient's cheek with a piece of tape until you can check for correct placement.	This prevents displacement of the NG tube while checking placement 
22. Verify tube placement according to agency policy.	The contents aspirated from the tube should be acidic with a pH <5. If the pH is more than 6, it may indicate the presence

Color-coded pH paper is usually used, as an initial and interim check, to confirm that acidic contents are present. Then an X-ray is taken to confirm placement prior to using NG tube for feeding.	of respiratory fluids or small bowel content, and the tube should be removed.
23. Once the tube placement has been confirmed, mark (with a permanent marker) and record the length of tubing extending from the nose to the outer end of the tube.	This aids in timely recognition and identification of tube displacement or migration.
24. Secure the tube to the patient's gown with a safety pin, allowing enough tube length for comfortable head movement.	
25. Document the procedure according to agency policy, and report any unexpected findings to the appropriate health care provider.	Timely and accurate documentation promotes patient safety

Tube feeding

Tube Feeding Preparation:

- Store the unopened products at room temperature, not in direct sunlight.
- Wash your hands prior to handling the products.
- Shake the can or bottle well and wipe the top clean prior to opening.
- Administer the feeding when the product is at room temperature.
- Cover any unused product and store in the refrigerator.
- Store the product for a maximum of 24 hours; if not used, discard the remainder.

Supplies to Gather:

- Correct tube feeding to be delivered (can or 24-hour bottle)
- Feeding pump (if on continuous tube feeding)
- Feeding set (bag set or spike set)
- 60 ml syringe -50 ml water -Graduated container

Positions for receiving tube feedings:

- Option 1: Propped up in bed or on a couch in a half sitting position with the head raised at least 30 degrees.
- Option 2: Sitting up in a chair.

Tube feeding procedure:

- wash hands.
- Assess abdomen for distention.
- Elevate head of the bed to at least 30 degrees.
- Insert 60 ml syringe into port and pour feeding product into syringe.
- Add formula to drain by gravity.
- Add more formula until the desired amount is infused.
- Flush with 50 ml of water.
- Cap the tube.

- Remain in upright position for at least one hour after feeding.
- Rinse equipment thoroughly.

Cleaning the Tube Site and Equipment:

- The site should be cleaned daily with soap and water.
- Do site care per the doctor's orders.
- Tube feeding bag sets may be cleaned with soap and water.
- Make sure to run the soapy water through the tubing as well.
- Tube feeding bag sets are good as long as they remain intact.
- Rinse with water to make sure all of the soap is out of the bag and the tubing.
- Syringes may be cleaned with soap and water or placed in a dishwasher on the top rack. They may also be used for as long as they remain intact.



When working with patient who have nasogastric tubes, remember the following care measures:

- Maintain and promote comfort. The tube constantly irritates the nasal mucosa, causing a great deal of discomfort. Ensure that the tube is securely anchored to the patient's nose to prevent excess tube movement, and is pinned to the gown to avoid excessive pulling or dragging.
- Because one nostril is blocked, patients tend to mouth breathe. This causes dehydration of the nasal and oral mucosa, and patients will complain of thirst, but they are usually NPO (nothing by mouth). Mouth care will help to relieve the dryness. This can include rinsing the mouth with cold water or mouthwash as long as the patient does not swallow. Some patients may be allowed to suck on ice chips.
- If the patient complains of abdominal pain, discomfort, or nausea, or begins to vomit, report it immediately. The drainage flow is probably obstructed and the tube will need to be irrigated.

Enema

Objectives:

Students will be able to:

- Define Enema.
- Identify types of Enema.
- Determine equipment's needed during the procedure.
- Applying Enema procedure.

Definition:

An enema is the injection of liquid into the rectum and colon by way of the anus.

Types of enema:

- Cleansing enemas: are given to remove feces from the colon.
- Oil-retention enemas: help to lubricate the stool and intestinal mucosa, making defecation easier.
- Carminative enemas: help to expel flatus from the rectum and relieve distention secondary to flatus.
- Medicated enemas: are used to administer a medication rectally.
- Anthelmintic enemas: are administered to destroy intestinal parasites.

Equipment:

- Enema solution (varies depending on reason for enema), often prepackaged, commercially prepared solutions
- Nonsterile gloves
- Waterproof pad
- Bath blanket
- Bedpan.
- Toilet tissue
- Water-soluble lubricant
- Additional PPE, as indicated
- Washcloth, skin cleanser, and towel

➤ Procedure

Steps	Rationale
1-Verify the order for the enema. Gather equipment.	Verifying the medical order is crucial to ensuring that the proper enema is administered to the right patient.
2- Perform hand hygiene and put on PPE, if indicated.	Hand hygiene and PPE prevent the spread of microorganisms.
3. Identify the patient.	Identifying the patient ensures the right patient receives the intervention and helps prevent errors.
4. Explain the procedure to the patient.	Explanation facilitates patient cooperation and reduces anxiety.
5. Assemble equipment on overbed table within reach. -Warm the enema solution to body temperature by placing the container in a bowl of warm water.	Arranging items nearby is convenient, saves time, and avoids unnecessary stretching and twisting of muscles on the part of the nurse. A cold solution can cause intestinal cramping.
6. Close the curtains around the bed and close the door to the room, if possible. Discuss where the patient will defecate. Have a bedpan, commode, or nearby bathroom ready for use.	This ensures the patient's privacy.
7. Adjust the bed to a comfortable working height, usually elbow height of the caregiver. Position the patient on the left side (Sims' position), as dictated by patient comfort and condition. Fold top linen back just enough to allow access to the patient's rectal area. Drape the patient with the bath blanket, as necessary, to maintain privacy and provide warmth. Place a waterproof pad under the patient's hip.	Sims' position facilitates flow of solution via gravity into the rectum and colon, optimizing retention of the solution.  Folding back the linen in this manner minimizes unnecessary exposure and promotes the patient's comfort and warmth. The waterproof pad will protect the bed.
8. Put on gloves.	Gloves prevent contact with blood and body fluids.
9. Remove cap of prepackaged enema solution. Apply a generous amount of lubricant to the tube.	Lubrication is necessary to minimize trauma on insertion.

10. Lift buttock to expose anus. Ask the patient to take several deep breaths. Slowly and gently insert the rectal tube 3 to 4 inches (7 to 10 cm) for an adult.	Good visualization of the anus helps prevent injury to tissues. Deep breathing helps relax the anal sphincters. 
11. Compress the container with your hands. Roll the end up on itself, toward the rectal tip. Administer all the solution in the container. Assess for dizziness, lightheadedness, nausea, diaphoresis, and clammy skin during administration.	Rolling the container aids administration of all of the contents of The container.
12. Remove the container while keeping it compressed. Have paper towel ready to receive tube as it is withdrawn.	If container is released, a vacuum will form, allowing some of the enema solution to re-enter the container.
13. Instruct the patient to retain the enema solution for at least 30 minutes or as indicated.	
14. Remove your gloves. Return the patient to a comfortable position. Make sure the linens under the patient are dry and ensure that the patient is covered.	
15. Raise side rail. Lower bed height and adjust head of bed to a comfortable position. Promotes patient safety.	
16. Remove additional PPE, if used. Perform hand hygiene.	
17. When the patient has a strong urge to dispel the solution, place him or her in a sitting position on bedpan or assist to commode or bathroom. Stay with the patient or have call bell readily accessible.	The sitting position is most natural and facilitates defecation. Fall prevention is a high priority due to the urgency of reaching the commode.
18. Remind the patient not to flush the commode before you inspect the results of enema, if used for bowel evacuation. Record character of stool, as appropriate, and the patient's reaction to the enema.	The results need to be observed and recorded.

19. Leave patient clean and comfortable. Care for equipment properly	Bacteria that grow in the intestine can be spread to others if equipment is not properly cleaned.
20. Perform hand hygiene.	Hand hygiene deters the spread of microorganisms.
21. Document the amount and type of enema solution used; length of time retained by the patient; amount, consistency, and color of stool, as appropriate; any abnormality.	

Colostomy care

Objectives:

Students will be able to:

- Define Colostomy.
- Determine equipment's needed during the procedure.
- Applying Enema procedure.

Definition: A colostomy is the creation of a stoma from part of the colon (large bowel), where the intestine is brought through the abdominal wall and attached to the skin, diverting normal intestinal fecal matter through the stoma instead of the anus.

Pouching Systems (Ostomy Appliances):

Individuals with colostomies have no control or sensation of frequency or output of the stoma. Patients with colostomy must wear a pouching system to collect the effluent from the stoma and protect the skin from irritation. The pouching system must be completely sealed to prevent leaking of the effluent and to protect the surrounding peristomal skin. The disposable pouching systems can be either a one-piece or a two-piece flexible system consisting of a plastic bag and a flange (skin barrier) that sit against the patient's skin. The flange may be flat or convex.

➤ **Colostomy care procedure**

Safety considerations:

- Pouching system should be changed every 4 to 7 days, depending on the patient and type of pouch.
- Always consult a wound care specialist or equivalent if there is skin breakdown, if the pouch leaks, or if there are other concerns related to the pouching system.
- Patients should participate in the care of their colostomy, and health care providers should promote patient and family involvement.
- Encourage the patient to empty the pouch when it is one-quarter to one-half full of feces, colostomy product choices are based on the patient's needs and preference.
- Follow all post-operative assessments for new ostomies according to agency policy.
- Medications and diet may need adjusting for new colostomies.
- A colostomy belt may be used to help hold the colostomy pouch in place.
- Factors that affect the pouching system include sweating, high heat, moist or oily skin, and physical exercise.
- Always treat minor skin irritations right away. Skin that is sore, wet, or red is difficult to seal with a flange for a proper leakproof fit.

➤ **procedure**

Steps	Rational
1. Perform hand hygiene.	This prevents the spread of microorganisms.
2. Gather supplies. 	Supplies include flange, ostomy bag and clip, scissors, stoma measuring guide, waterproof pad, pencil, adhesive remover for skin, powder, wet cloth, non-sterile gloves, and additional cloths.
3. Identify the patient and review the procedure. Encourage the patient to participate as much as possible or observe/assist patient as they complete the procedure.	Encouraging patients to participate helps them adjust to having an ostomy
4. Create privacy. Place waterproof pad under pouch.	The pad prevents the spilling of effluent on patient and bedsheets.
5. Apply gloves. Remove ostomy bag, and measure and empty contents. Place old pouching system in garbage bag 	
6. Remove flange by gently pulling it toward the stoma. Support the skin with your other hand. An adhesive remover may be used 	Gentle removal helps prevent skin tears. An adhesive remover may be used to decrease skin and hair stripping.

7. Clean stoma gently by wiping with warm water. Do not use soap.



Aggressive cleaning can cause bleeding. If removing stoma adhesive paste from skin, use a dry cloth first.

8. Assess stoma and peristomal skin.



A stoma should be pink to red in colour, raised above skin level, and moist.

9. Measure the stoma diameter using the measuring guide (tracing template) and cut out stoma hole.

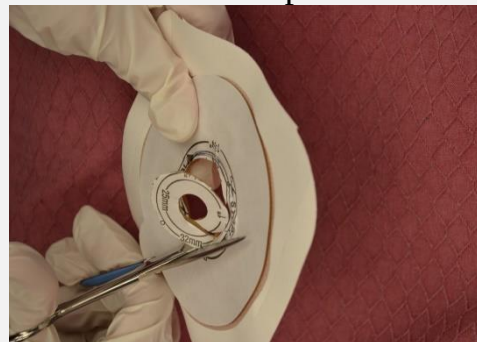
Trace diameter of the measuring guide onto the flange, and cut on the outside of the pen marking

The opening should be 2 mm larger than the stoma size.

Keep the measurement guide with patient supplies for future use



Trace template



Once size is traced onto back of flange, cut out size to fit stoma



Assess flange for proper fit to stoma

- 9.** Prepare skin and apply accessory products as required or according to agency policy.



Wet skin will prevent the flange from adhering to the skin.

- 10.** Remove inner backing on flange and apply flange over stoma. Leave the border tape on. Apply pressure. Hold in place for 1 minute to warm the flange to meld to patient's body. Then remove outer border backing and press gently to create seal.



The warmth of the hand can help the appliance adhere to the skin and prevent leakage.



11.Apply the ostomy bag. Attach the clip to the bottom of the bag.

This step prevents the effluent from soiling the patient or bed.



13. Hold palm of hand over ostomy pouch for 2 minutes to assist with appliance adhering to skin.

The flange is heat activated.

14. Clean up supplies, and place patient in a comfortable position. Remove garbage from patient's room.

Removing garbage helps decrease odor.

15. Perform hand hygiene.

This minimizes the transmission of microorganisms.

16. Document procedure.

Document appearance of stoma and peristomal skin, products used, and patient's ability to tolerate procedure and assistance with procedure.

Routine catheter care

Objectives:

Students will be able to:

- Determine equipment's needed during the procedure.
- Applying Routine catheter care.

Over view of the skill

- An indwelling catheter is used to drain urine from the bladder but may provide a route for infection to enter the body. Therefore, care must be taken to ensure that the surrounding area is clean to decrease contamination of the catheter by bacterial flora.
- **Goal:** The patient's skin remains clean, dry, intact, and without evidence of irritation or breakdown; and the patient verbalizes an understanding of the purpose for, and care of the catheter, as appropriate

➤ Equipment

- Prepackaged kit
- Antiseptic solution
- Sterile swabs
- Clean gloves
- Sterile bowl
- Wash cloth, soap, and water
- Chlorhexidine 2% cloth (*optional*)

➤ Procedure

Steps	Rational
1. Wash hands.	Reduces the transmission of microorganisms.
2. Check agency protocol or care plan.	Ensures proper procedure.
3. Identify client and explain procedure.	Prepares client.
4. Provide privacy.	Protects client dignity.
5. Position patient with waterproof pad under buttocks and cover with bath blanket, exposing only genital area and catheter. a Female in dorsal recumbent position. b Male in supine position.	Allows for visualization of field. If unable to visualize the perineal area with the client supine, try placing the client in a side lying position.
6. Put on clean gloves.	Reduces transmission of microorganisms.
7. Cleanse perineal area with soap and water. Option: Use chlorhexidine 2% cloth	Soap has antibacterial qualities adequate to clean the area and will not usually irritate the skin or mucous membranes.

8. Catheter care:

a. Female: Use nondominant hand to gently separate labia to fully expose urethral meatus and catheter. Maintain position of hand throughout procedure.

b. Male: Use nondominant hand to retract foreskin if not circumcised and hold penis at shaft just below glans. Maintain hand position throughout procedure



Provides full visualization of urethral meatus. Full separation of labia prevents contamination of meatus during cleaning.

Retraction of foreskin provides full visualization of urethral meatus.

9. Cleanse catheter from meatus out to end of catheter, taking care not to pull on catheter.



Moving from most clean area out does not traumatize urethra or bladder.

10. Be sure to repeat catheter care anytime it becomes soiled with stool or other drainage.

Prevents infection.

11. Place linen or cotton balls in proper receptacle for laundry or disposal.

Reduces transmission of infection to other clients.

12. Wash hands.

Reduces transmission of microorganisms.

➤ Procedure Check List

PROCEDURE STEPS	Mark	Trial 1	Trial 2	Trial 3	comment
1. Wash hands.					
2. Check agency protocol or care plan.					
3. Identify client and explain procedure.					
4. Provide privacy.					
5. Position patient with waterproof pad under buttocks and cover with bath blanket, exposing only genital area and catheter. a Female in dorsal recumbent position. b Male in supine position.					
6. Put on clean gloves.					
7. Cleanse perineal area with soap and water. Option: Use chlorhexidine 2% cloth					
8. Catheter care: a. Female: Use nondominant hand to gently separate labia to fully expose urethral meatus and catheter. Maintain position of hand throughout procedure. b. Male: Use nondominant hand to retract foreskin if not circumcised and hold penis at shaft just below glans. Maintain hand position throughout procedure					
9. Cleanse catheter from meatus out to end of catheter, taking care not to pull on catheter.					
10. Be sure to repeat catheter care anytime it becomes soiled with stool or other drainage.					
11. Place linen or cotton balls in proper receptacle for laundry or disposal.					
12. Wash hands.					

Glasgow Coma Scale (GCS)

Objectives:

Students will be able to:

- Define Glasgow Coma Scale (GCS).
- Identify The goal of Obtaining Glasgow Coma Scale.
- Explain Glasgow Coma Scale (GCS) scoring system.
- Demonstrate Glasgow Coma Scale (GCS) assessment.

➤ Definition

- The Glasgow Coma Scale (GCS) is a neurological assessment scale that evaluates three key categories of behavior that most closely reflect activity in the higher centers of the brain: **eye opening, verbal response, and motor response**. Within each category, each level of response is given a numerical value.

➤ The goal of Obtaining Glasgow Coma Scale

- It is commonly used in the context of head trauma, but it is also useful in a wide variety of other non-trauma related settings.
- Regular assessment of a patient's GCS can identify early signs of neuroglial deterioration.
- The GCS is used in conjunction with other neurologic assessments, including pupillary reaction and vital sign measurement, to evaluate a patient's neurological status.

➤ Glasgow Coma Scale (GCS) content and scoring system.

- The GCS is a more accurate evaluation of mental status over time. Some limitations to using the GCS include the inability to assess the verbal score after the patient has been intubated.
- There are three aspects of behavior that are independently measured as part of an assessment of a patient's GCS
 - ✓ **eye-opening** 4 points
 - ✓ **verbal response.** 5 points
 - ✓ **motor response** 6 points
- Total score classified as: Patients, who open their eyes spontaneously, obey commands (a fully awake, alert, and oriented patient) score a total of **(15) points**, the best possible score, whereas flaccid patients, who neither open their eyes or verbalize, score the **minimum of (3) points**.

Component	Score	Response
Eye opening	4	Opens eyes spontaneously when someone approaches
	3	Opens eyes in response to speech (normal tone or shouting)
	2	Opens eyes only to painful stimuli (apply pressure with a pen to the lateral outer aspect of the second or third finger, up to 10 seconds, then release)
	1	No response to painful stimuli
Best verbal response	5	Converses; oriented to time, place, and person
	4	Converses; disoriented to time, place, or person; any one or all indicators
	3	Converses only in words or phrases that make little sense in the context of the questions
	2	Responds with incomprehensible sounds; no understandable words and/or moaning, groaning, or crying in response to painful stimuli
	1	No response
Best motor response	6	Accurately responds to instructions; obeys a simple command, such as "Lift your left hand off the bed"
	5	Localizes (moves hand to point of stimulation) to painful stimuli and attempts to remove source
	4	Flexion reflex action, but unable to locate the source of pain; purposeless movement in response to pain
	3	Flexes elbows and wrists while extending lower legs to pain; decorticate posturing
	2	Extends upper and lower extremities to pain; decerebrate posturing
	1	No motor response to pain on any limb

Item	Adult	1-5 years	0-1 years
Eye Opening			
4	Spontaneously	Spontaneously	Spontaneously
3	To command	To command	To command
2	To pain	To pain	To pain
1	No response	No response	No response
patient score			
Verbal Response			
5	Oriented	appropriate words, phrases	coos, babbles, smiles
4	Confused	inappropriate words	cries

3	Inappropriate words	cries, screams	inappropriate cries, screams
2	Incomprehensible	grunts	grunts
1	No response	no response	no response
patient score			
Motor Response			
6	Obeys commands	spontaneous	spontaneous
5	Localizes pain	localizes pain	localizes pain
4	Withdraws from pain	flexion withdrawal	flexion withdrawal
3	Abnormal flexion	abnormal flexion	abnormal flexion
2	Extension	extension	extension
1	No response	no response	no response
patient score			
Total patient score			

- A GCS score of (11 to 13) points indicates **Lethargy**, score of (8 to 10) points indicates **stupor**, a score of less than 7 indicates **a coma** and score of 3 indicate **deep coma**

Levels of Consciousness

Term	Responsiveness
Alert	Normal. Responds with minimal stimuli.
Awake	Somewhat confused on awakening, but fully oriented.
Confused	Oriented to person, but not time/place. Decreased attention span.
Delirious	Disoriented to time, place, and person. May have visual and auditory hallucinations.
Lethargic	Drowsy but follows simple commands. Requires increased stimulation to respond.
Obtunded	Indifferent to external stimulation. Responds with a word or two.
Stuporous	Responds to vigorous and continuous external stimuli. Limited spontaneous movement.
Comatose	Reflexive posturing or no response to any stimulus.

- A GCS score of 13-14 points is associated with **mild head injury**, score of 8-12 points is associated with **moderate head injury** and score of less than 8 is associated with **severe head injury**

➤ **Demonstrate Glasgow Coma Scale (GCS) assessment.**

Procedure steps	Mark	Trial 1	Trial 2	Trial 3	comment
1. Performed hand washing technique using the principle of 5 moments of hand hygiene					
2. Correctly verified patient identification using two identifiers.					
3. Obtained consent and be able to explain to patient/ sitter about the procedure.					
4. Ensured patient privacy during the procedure.					
5. Explain the results of the GCS using EVM (eye opening, verbal response and motor response) and its implications of abnormal results.					
6. Use the communication tool if necessary (for any language barrier)					
7. Eye opening: - Observe if eyes open spontaneously. - Call patient's name twice if patient is not opening his eyes. - Apply peripheral pain stimulus (pressing the lateral part of the nail bed)					
8. Verbal response - Check patient's orientation by asking (time, place and person)					
9. Motor response: - Check patient if obeys commands by instructing: (e.g., squeezing your hand, close/ open eyes - If a patient is not obeying command, apply the central pain stimulus (trapezius squeeze) - Observe if patient reacts to pain: (i.e.: localizing, withdrawing, abnormal flexion and extension)					
10. Have a torch for patient assessment of pupillary reaction. - Check pupil size before the reaction to light and able to differentiate pupil reactions: (ie, brisk, sluggish, or fixed)					
11. Correctly perform assessment of limb movements and able to differentiate: - Normal power; Mild weakness-; Severe weakness; No movement; Abnormal posture (extension/ flexion)					
12. Verbalized and documented correctly - Summarize, interpret and report abnormal results of the GCS.					

Adapted from Catanguí, E. (2019). Improving Glasgow Coma Scale (GCS) competency of nurses in one acute stroke unit-a nursing initiative project. J Nurs Pract, 3(1), 109-15.

Caring For a Patient in Skeletal Traction

Objectives:

Students will be able to:

- Define Skeletal Traction.
- Identify The goal of applying Traction.
- Enumerate the types of Traction and skin traction.
- Demonstrate Caring For a Patient in Skeletal Traction procedure.

➤ Definition of Traction

- Means that is an application of a pulling force to a part of the body or an extremity while a counter traction pulls in opposite direction.

➤ Definition of Skeletal traction

- Skeletal traction provides pull to a body part by attaching weight directly to the bone, using pins, screws, wires, or tongs (skull traction). It is used to immobilize a body part for prolonged periods. This method of traction is used to treat fractures of the femur, tibia, and cervical spine.

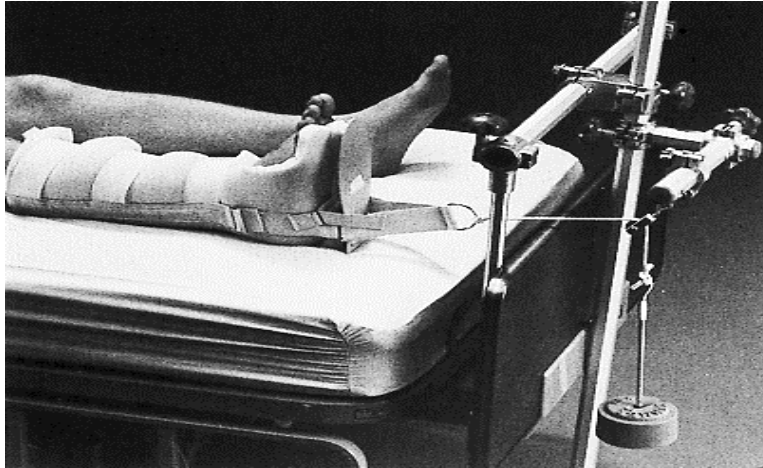
➤ The goal of applying Traction.

- To secure immobilization of fracture parts.
- To minimize or reduce muscle spasms.
- To relieve pain.
- To realign bones.
- To prevent or correct deformity and
- To increase space between opposing surfaces

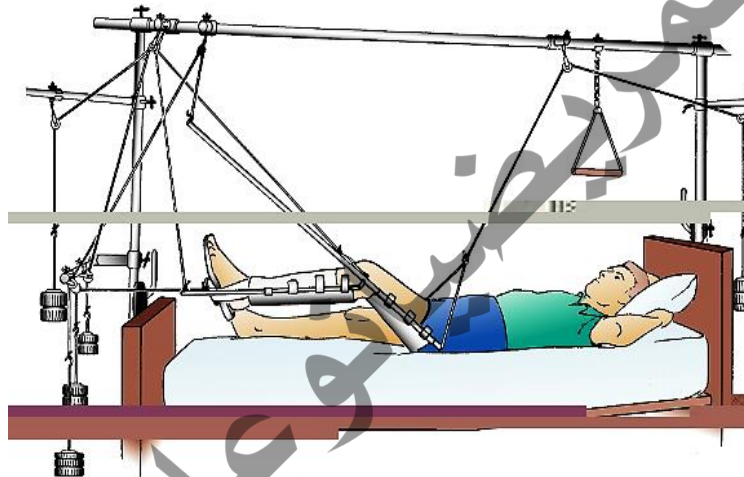
➤ Types of traction: -

- Straight or running traction.
- Suspension or balanced traction.
- Manual traction.
- Skeletal traction
- Skin traction.

- ***Straight or running traction.*** Applies the pulling force in a straight line with the body part resting in the bed.



- ***Suspension or balanced traction.*** supports the affected extremity off the bed and allow for some patient movement without disruption of the line of pull



- ***Skin traction.*** is the application of a pulling force to the skin and soft tissues, the material used including wrapping bandage, tapes, slings and the tapes.



- **Skeletal traction** Is traction applied directly to the skeletal system with such device as pins, wires and tongs into the bones continuously for extended period.



- **Manual traction** Means pulling on the body using a person's hand and muscular strength. It is most often used briefly to realign broken bones. It is used to replace a dislocated bone into its original position within a joint

➤ **Principles of effective traction:**

- Counter traction must be used to achieve effective traction.
- Traction must be continuous to be effective in reducing & immobilizing fractures.
- Skeletal traction is never interrupt.
- Weights are not removed unless intermittent traction is prescribed.
- Any factor that might reduce the effective pull or alter its resultant line of pull must be eliminated:
- The patient must be in a good body alignment in center of the bed when traction applied.
- Ropes must be UN obstructed.
- Weight must be hanged freely & not rest on the bed or floor.
- Knots in the ropes or the footplate must not touch the pulley or the foot of the bed

➤ **Complication of traction:**

- Skin breakdown,
- Nerve pressure &
- Circulatory impairment.

➤ **Complication of skull traction**

- Infection.
- Skin breaks down.
- Application of splint difficult.
- Distraction at fracture site.
- Ligament damage.
- Physical damage.
- Depressed scars.

➤ **Nursing intervention:**

1. Maintaining Effective Traction.
2. Maintaining body alignment.
3. Preventing Skin breakdown.
4. Preventing Circulatory impairments.
5. Monitoring Neurological Status.
6. Providing Pin Site Care.
7. Promoting exercise

➤ **Procedure**

Steps	Rational
1. Review the medical record and the nursing plan of care to determine the type of traction being used and the prescribed care.	Reviewing the medical record and plan of care validates the correct patient and correct procedure.
2. Perform hand hygiene. Put on PPE, as indicated.	Hand hygiene and PPE prevent the spread of microorganisms. PPE is required based on transmission precautions.
3. Identify the patient. Explain the procedure to the patient, emphasizing the importance of maintaining counterbalance, alignment, and position	Patient identification validates the correct patient and correct procedure. Discussion and explanation help allay anxiety and prepare the patient for what to expect.
4. Close the curtains around the bed and close the door to the room, if possible	Closing the door or curtains provides for privacy.
5. Perform a pain assessment. Administer prescribed medications in sufficient time to allow for the full effect of the analgesic and/or muscle relaxant	Assessing for pain and administering analgesics promote patient comfort.
6. Ensure the traction apparatus is attached securely to the bed. Be sure that the weights hang freely, not touching the bed or the floor.	Proper traction application reduces the risk of injury by promoting accurate counterbalance and traction function
7. Check the alignment of the patient's body, as prescribed.	Proper alignment maintains an effective line of pull and prevents injury.
8. Assess the site at and around the pins for redness, edema, and odor. Assess for skin tenting, prolonged or purulent drainage, elevated body temperature, elevated pin site temperature, and bowing or bending of the pins.	Pin sites provide a possible entry for microorganisms. Skin inspection allows for early detection and prompt intervention should complications develop.

9. Provide pin-site care. a. Using sterile technique. b. Put on the sterile gloves c. Clean the pin site, starting at the insertion area and working outward, away from the pin site 	Cleaning from the center outward ensures movement from the least to most contaminated area.
10. Depending on medical order and facility policy, apply the antimicrobial ointment to pin sites and apply a dressing.	Antimicrobial ointment helps reduce the risk of infection. A dressing aids in protecting the pin sites from contamination and in containing any drainage.
11. Place the bed in the lowest position that still allows the weight to hang freely.	Proper bed positioning ensures effective application of traction without patient injury
12. Remove PPE, if used. Perform hand hygiene	Removing PPE properly reduces the risk for infection transmission and contamination of other items. Hand hygiene prevents the spread of microorganisms.

➤ Procedure Check List

Procedure Steps	Mark	Trial 1	Trial 2	Trial 3	comment
1. Review the medical record and the nursing plan of care.					
2. Perform hand hygiene. Put on PPE, as indicated.					
3. Identify the patient. Explain the procedure to the patient.					
4. Close the curtains around the bed and close the door to the room, if possible					
5. Perform a pain assessment. Administer analgesic and/or muscle relaxant					
6. Ensure the traction apparatus is attached securely to the bed. Be sure that the weights hang freely, not touching the bed or the floor.					
7. Check the alignment of the patient's body, as prescribed.					
8. Assess the site at and around the pins for redness, edema, and odor. Assess for skin tenting, prolonged or purulent drainage, elevated body temperature, elevated pin site temperature, and bowing or bending of the pins.					
9. Provide pin-site care. a. Using sterile technique. b. Put on the sterile gloves c. Clean the pin site, starting at the insertion area and working outward, away from the pin site					
10. apply the antimicrobial ointment to pin sites and apply a dressing as prescribed.					
11. Place the bed in the lowest position that still allows the weight to hang freely.					
12. Remove PPE, if used. Perform hand hygiene					

Caring For a Cast

Objectives:

Students will be able to:

- Define a cast.
- Identify The goal of applying a cast.
- Demonstrate Caring for a Cast procedure.

➤ Definition

- A cast is a rigid external immobilizing device that encases a body part. Casts, made of plaster or synthetic materials, such as fiberglass, are used to immobilize a body part in a specific position and to apply uniform pressure on the encased soft tissue.

➤ The goal of applying a cast.

- Immobilize the injured structure.
- Apply for fracture.
- Correct deformity.
- Provide support and stability for weakened joints.

➤ Nursing care after cast application

- Regular assessment of the patient to prevent complication or detect their early signs so that prompt treatment can be instituted and assess emotional condition.
- Assessment for adequate sensation & circulation especially capillary refills.
- Pain and pressure the nurse should not ignore any of complaints of pain & pressure these may be due to compression of blood vessels & nerves.
- Document & report any signs of pressure under the cast if pressure is not relieved the necrotic area may drain, stain the cast and emit an odor (offensive smell).
- Symptoms of compartment syndrome remember that all begin with “p” pain, paralysis & pulselessness.
- Neurovascular assessment: In all casted patients, color, motion, temperature and sensation of finger or toes should be observed every 30 minutes for several hours after cast application. (Every 3 hours if there is edema).
- Circulatory impairment result in symptoms of coldness, pain and numbness in toes or fingers.
- Normal findings: minimal edema, minimal discomfort, pink color, warm to touch, rapid capillary refill response (1: 5 sec), normal sensation and ability to exercise the fingers or toes

➤ Procedure

Steps	Rational
1. Review the medical record and the nursing plan of care to determine the type of traction being used and the prescribed care.	Reviewing the medical record and plan of care validates the correct patient and correct procedure.
2. Perform hand hygiene. Put on PPE, as indicated.	Hand hygiene and PPE prevent the spread of microorganisms. PPE is required based on transmission precautions.
3. Identify the patient. Explain the procedure to the patient, emphasizing the importance of maintaining counterbalance, alignment, and position	Patient identification validates the correct patient and correct procedure. Discussion and explanation help allay anxiety and prepare the patient for what to expect.
4. Close the curtains around the bed and close the door to the room, if possible	Closing the door or curtains provides for privacy.
5. If the cast is on an extremity, elevate the affected area on pillows covered with waterproof pads. Maintain the normal curvatures and angles of the cast	Elevation helps reduce edema and enhances venous return. Use of a waterproof pad prevents soiling of linen. Maintaining curvatures and angles maintains proper joint alignment, helps prevent flattened areas on the cast as it dries, and prevents pressure areas.



6. Assess the condition of the cast.



Be alert for cracks, dents, or the presence of drainage from the cast. Perform skin and neurovascular assessments according to facility policy, as often as every 1 to 2 hours. Check for pain, edema, inability to move body parts distal to the cast, pallor, pulses, and abnormal sensations. If the cast is on an extremity, compare it with the noncasted extremity



Assessment helps detect abnormal neurovascular function or infection and allows for prompt intervention. Assessing the neurovascular status determines the circulation and oxygenation of tissues. Pressure within a cast may increase with edema and lead to compartment syndrome

7. If breakthrough bleeding or drainage is noted on the cast, mark the area on the cast, according to facility policy. Indicate the date and time next to the area.



Marking the area provides a baseline for monitoring the amount of bleeding or drainage.

8. Assess for signs of infection. Monitor the patient's temperature. Assess for a foul odor from the cast, increased pain, or extreme warmth over an area of the cast.

Infection deters healing. Assessment allows for early detection and prompt intervention

9. Reposition the patient every 2 hours. Provide back and skin care frequently. Encourage range of motion exercises for unaffected joints. Encourage the patient to cough and breathe deeply.

Repositioning promotes even drying of the cast and reduces the risk for the development of pressure areas under the cast. Frequent skin and back care prevent patient discomfort and skin breakdown.

10. Place the bed in the lowest position, with the side rails up.

Having the bed at proper height ensures patient safety.

11. Remove PPE, if used. Perform hand hygiene.

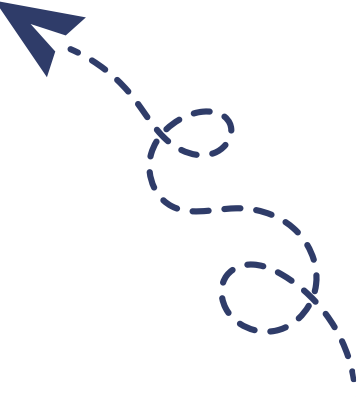
Removing PPE properly reduces the risk for infection transmission and contamination of other items. Hand hygiene prevents the spread of microorganisms.

➤ Procedure Check List

Procedure Steps	Mark	Trial 1	Trial 2	Trial 3	comment
1. Review the medical record and the nursing plan of care.					
2. Perform hand hygiene. Put on PPE, as indicated.					
3. Identify the patient. Explain the procedure to the patient.					
4. Close the curtains around the bed and close the door to the room, if possible					
5. Elevate the affected area on pillows covered with waterproof pads. Maintain the normal curvatures and angles of the cast					
6. Assess the condition of the cast. Be alert for cracks, dents, or the presence of drainage from the cast. Perform skin and neurovascular assessments according to facility policy, as often as every 1 to 2 hours. Check for pain, edema, inability to move body parts distal to the cast, pallor, pulses, and abnormal sensations. If the cast is on an extremity, compare it with the noncasted extremity					
7. If breakthrough bleeding or drainage is noted on the cast, mark the area on the cast, according to facility policy. Indicate the date and time next to the area.					
8. Assess for signs of infection. Monitor the patient's (temperature, a foul odor from the cast, increased pain, or extreme warmth over an area of the cast).					
9. Reposition the patient every 2 hours. Provide back and skin care frequently. Encourage range of motion exercises for unaffected joints. Encourage the patient to cough and breathe deeply.					
10. Place the bed in the lowest position, with the side rails up.					
11. Remove PPE, if used. Perform hand hygiene.					

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يُرجى زيارة قناة ثانية تمريضيانو
على تطبيق التليجرام



للاضمام إلى القناة، يُرجى النقر على زر subscribe